
Science and
Technology
Facilities Council
ISIS Neutron and
Muon Source
Welcome

Work Experience 2026
Introduction

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RAL Visitor Centre

20.07.26



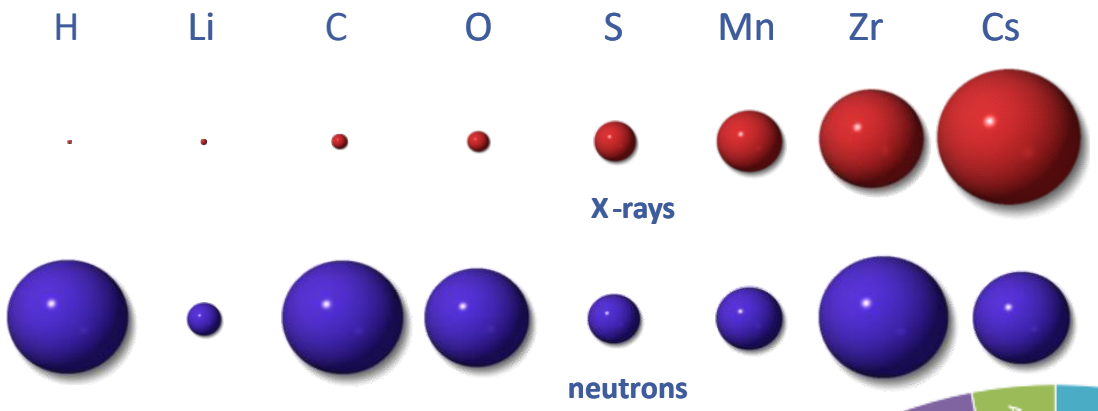
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2. What ISIS Does & how you will help
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6. Getting started

Intro & Welcome


Welcome to ISIS

What does ISIS do?



NEUTRON	
diffraction	Neutron diffraction experiments determine the atomic and/or magnetic structure of a material
spectroscopy	Neutron spectroscopy measures the atomic and magnetic motions of atoms
imaging	Radiography and tomography to a unique view of materials
reflectometry	Neutron reflectometry is a technique for measuring the structure of thin films and interfaces
small angle	Small angle neutron scattering is able to probe structures at length scales from around 1 nm to more than 100 nm





27 Graduates

49 Co-sponsored PhD students

2 Vacation students

17 Apprentice placements

32 Undergraduate sandwich students

Total number of visitors **3446**

2668 School students, teachers and public visitors

603 Business and government visitors

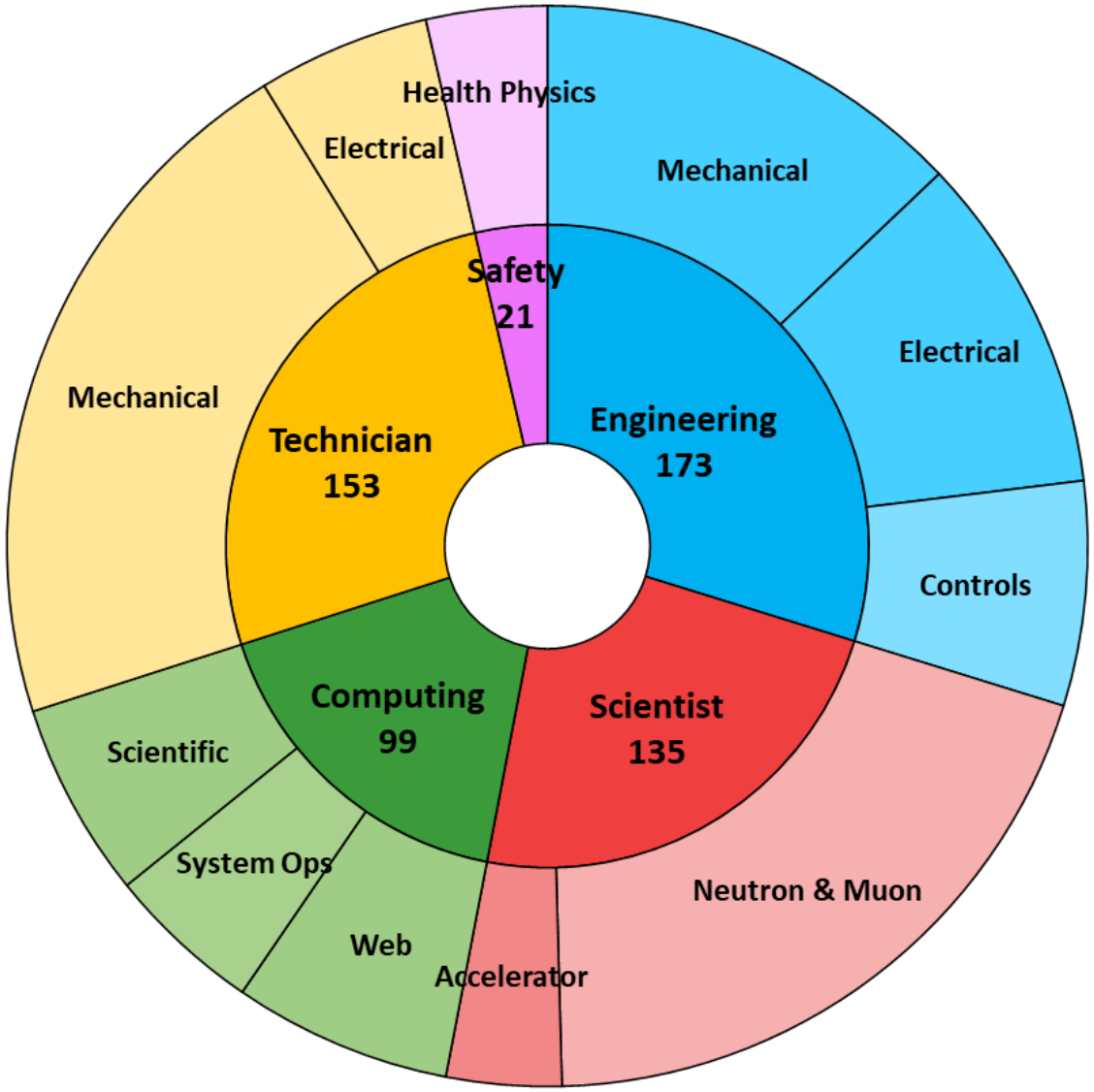
175 Higher education

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What does ISIS do? **Demographics**





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Industrious Neutrons

NVIDIA's advanced commercial 'off-the-shelf' GPUs have become a key technology for AI and Machine Learning applications including fully autonomous cars. Understanding GPU resilience at ChipIr has become a priority for such safety critical applications.

"The ChipIR beamline has been extremely valuable for our studies into the effects of soft errors on NVIDIA GPUs. The stability and strength of the beam and the well-thought-out infrastructure have made it possible to run more experiments in less time."

Mike Sullivan, NVIDIA







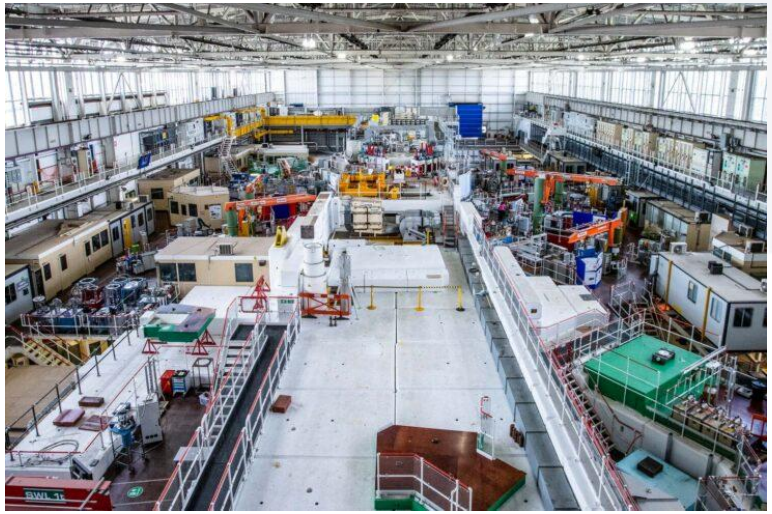
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How do we produce neutrons and muons?











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Quiz

- What do these acronyms mean (and what do they do)?:
  - STFC?
  - RAL?
  - ISIS?



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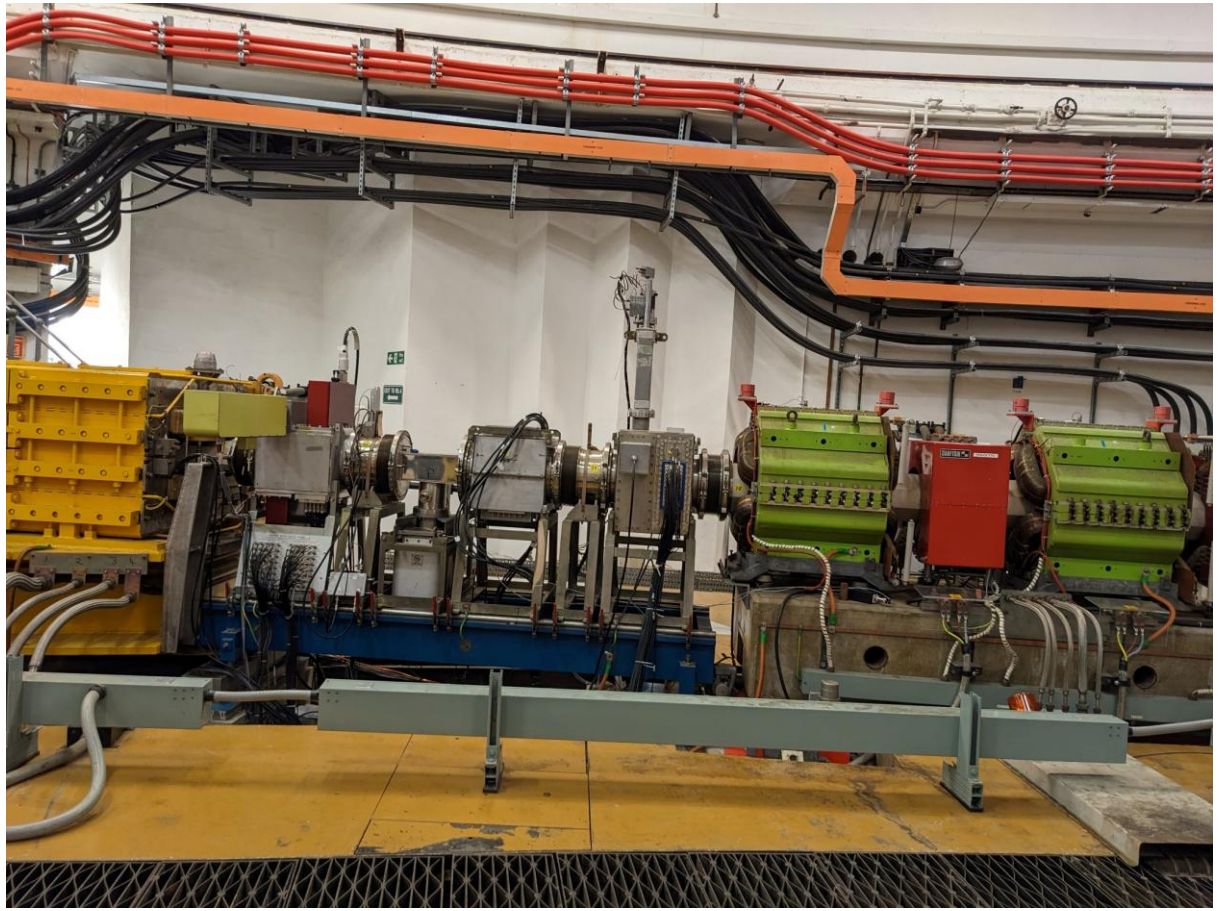
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ISIS Accelerators

Reference for you to look at in your own time



Accelerator Physics



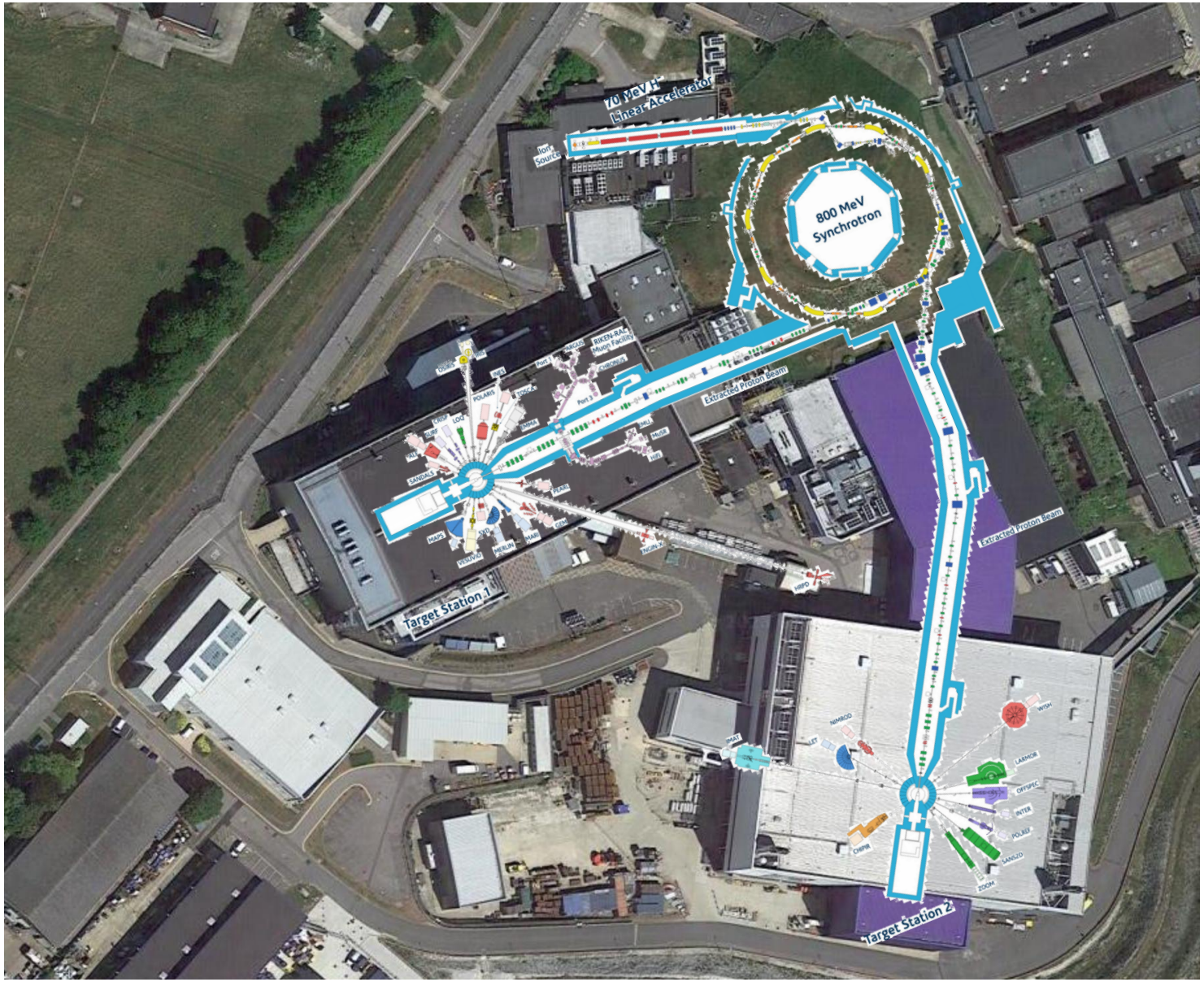
We will be looking at:

1. **Closed Orbit**: The beam path or position as it orbits the synchrotron. This is controlled by ‘Orbit Corrector’ dipole magnets.

2. **Tune**: related to the beam envelope, or focussing as it orbits the synchrotron. This is controlled by quadrupoles, which are focussing magnets.

We will split into teams, each with a specific goal. We will also work as a single team to bring together the workflows into a coherent tool.

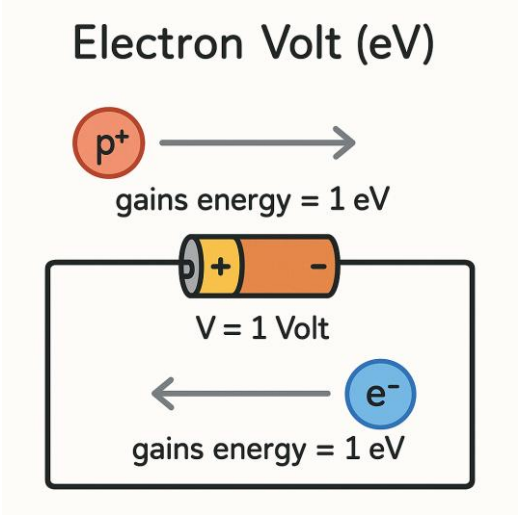
How do we produce neutrons and muons?



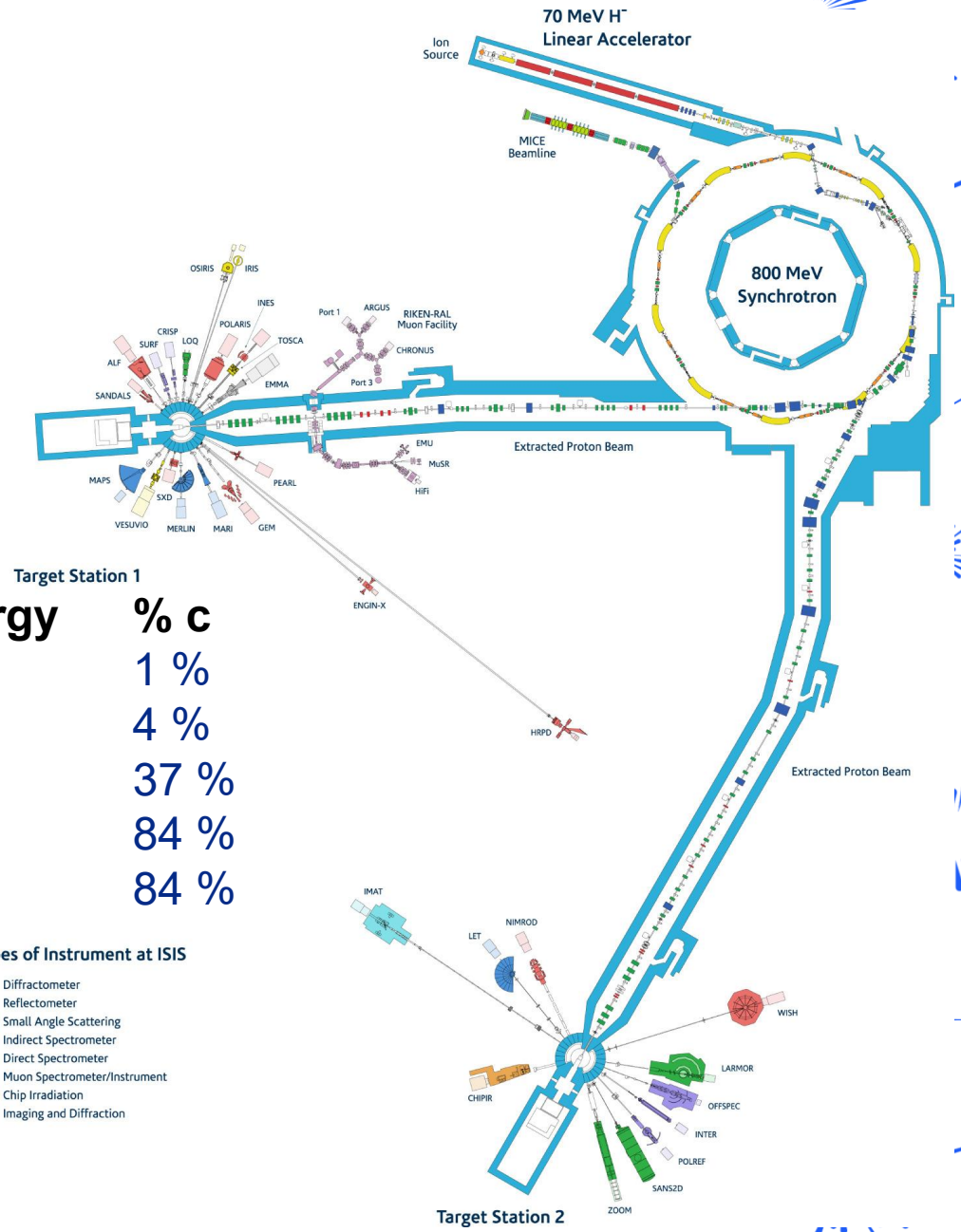
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How do we produce neutrons and muons?
Particle Accelerators



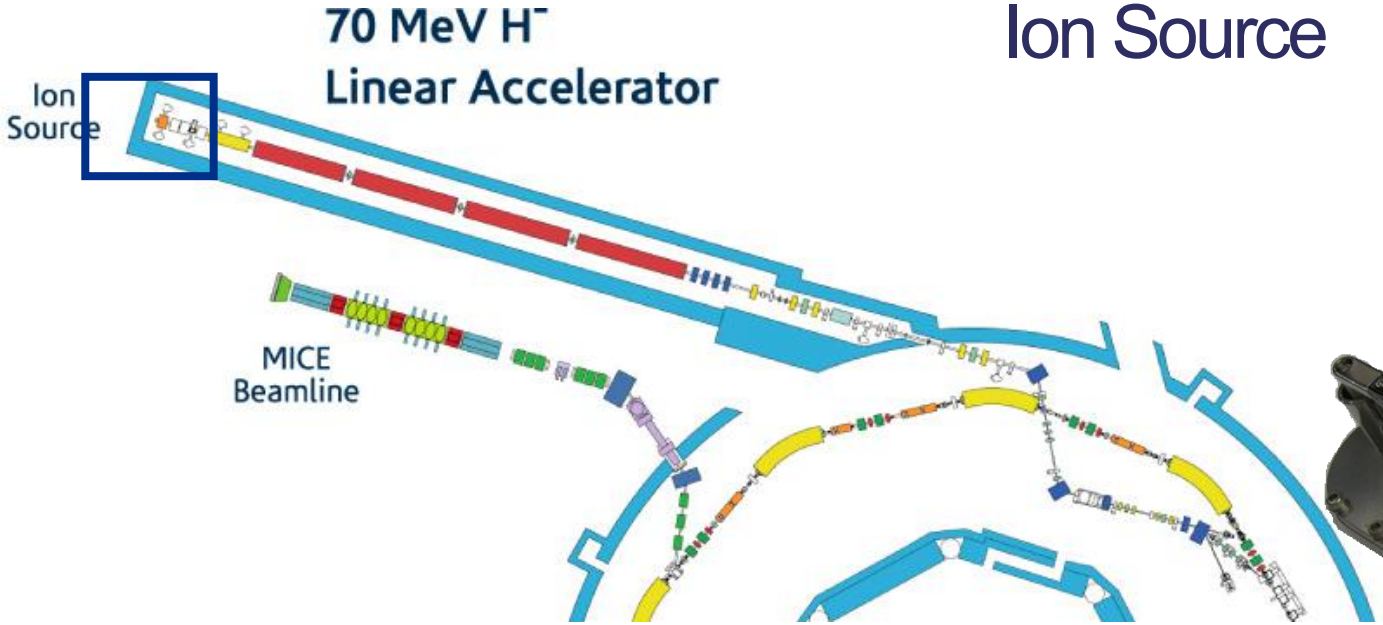
• Ion	Accelerator / Transfer line	Beam Energy	% c
• H^-	Ion Source	35 keV	1 %
• H^-	Radio Frequency Quadrupole	665 keV	4 %
• H^-	Linear Accelerator x 4	70.4 MeV	37 %
• Proton	Synchrotron	800 MeV	84 %
• Proton	Extraction Lines x 2	800 MeV	84 %

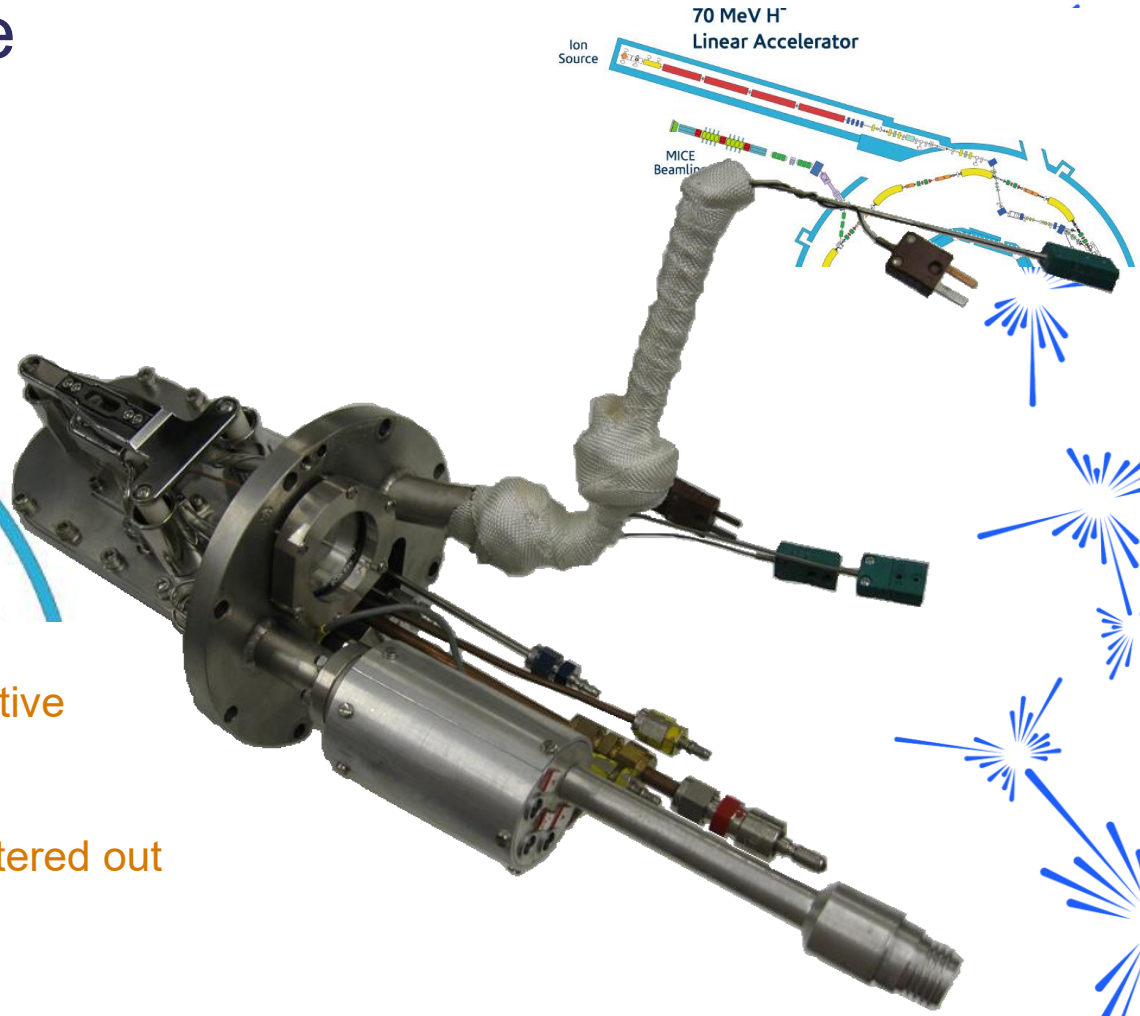


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Ion Source





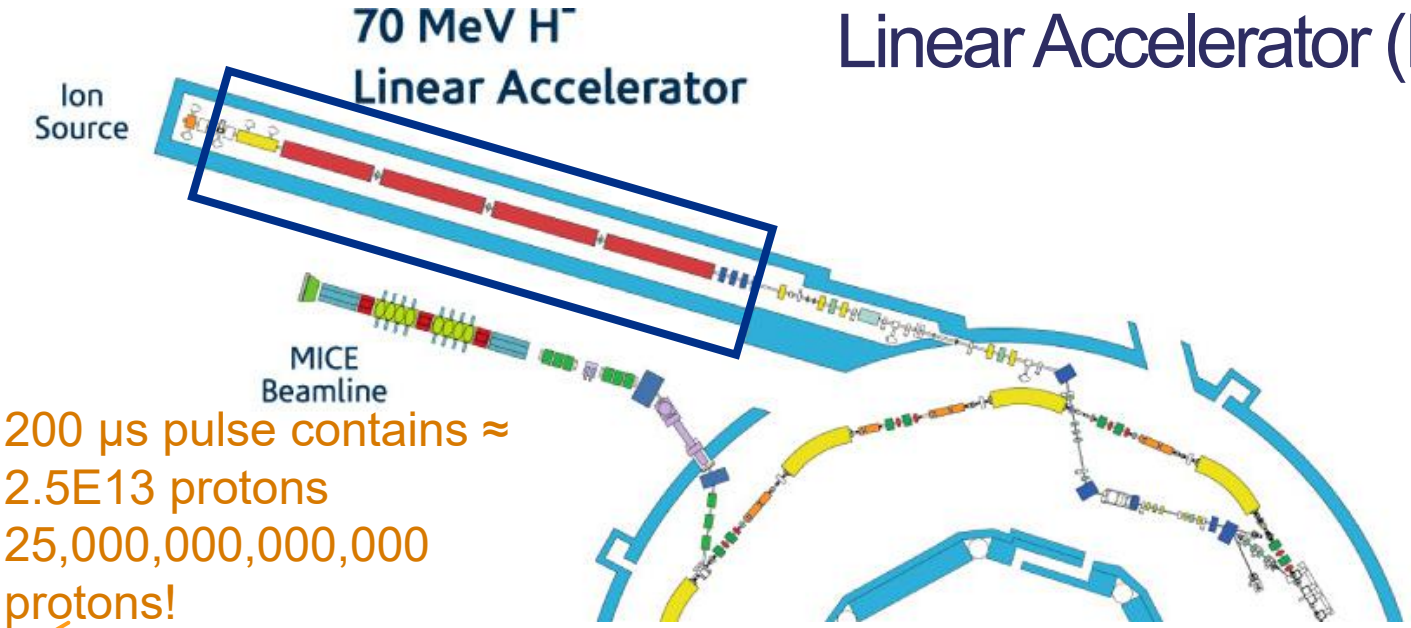
Hydrogen gas is combined with a hot caesium vapour to produce negative hydrogen ions (proton + 2 electrons)

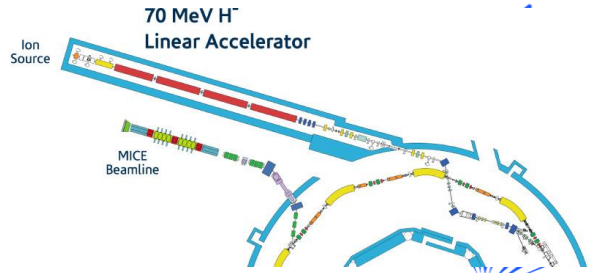
These are directed using strong electric fields and free electrons are filtered out using a magnetic field.

H^- ions extracted at 35 keV in **~200 µs pulses @ 50 Hz.**

The plasma ultimately damages components, so the source is periodically replaced.

Linear Accelerator (Linac)





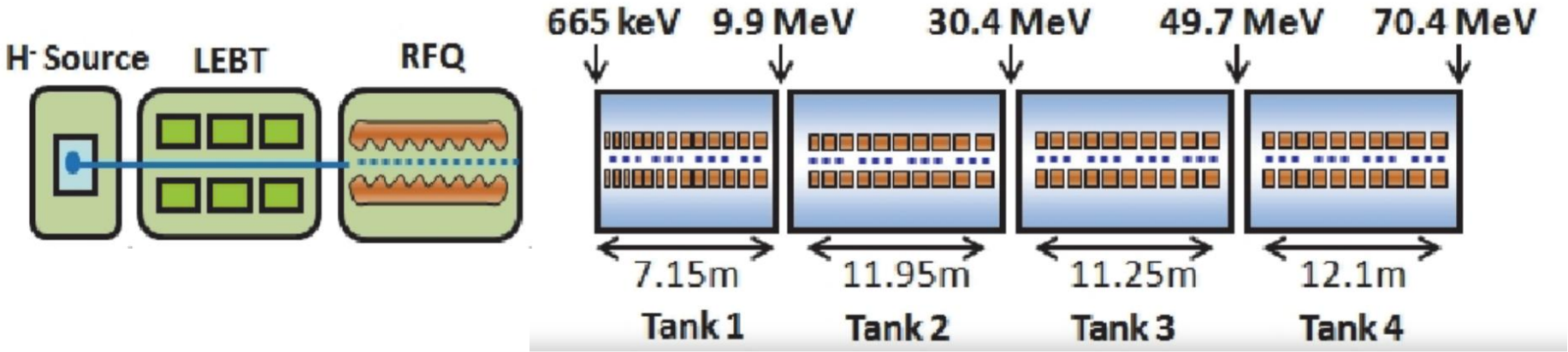
- Few mm spot size on injection foil

200 µs pulse contains ≈
2.5E13 protons
25,000,000,000,000
protons!

≈ 200 µs @ 70 MeV
≈ 22,000 m @ 37% c
≈ 248,000,000 mph

Perpendicular to direction of travel = transverse (x, y)

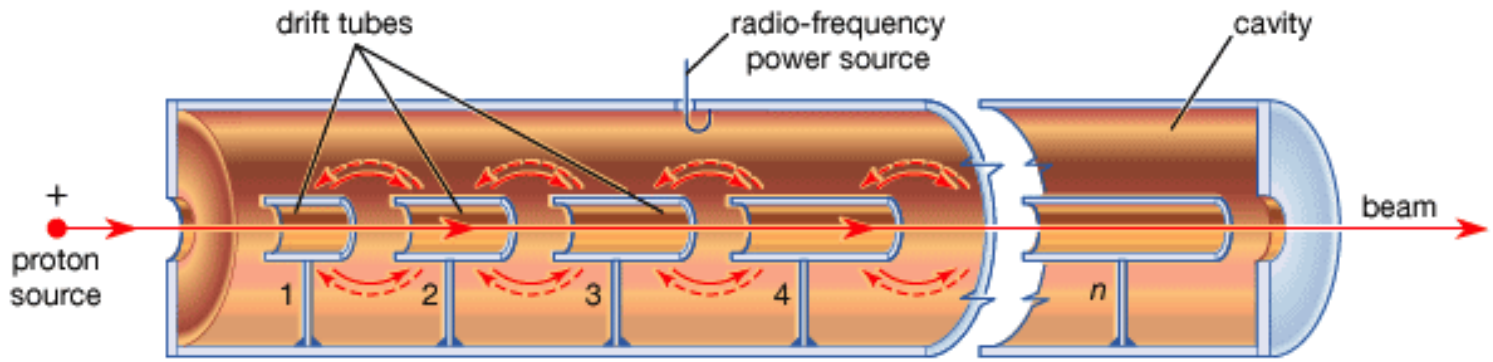
H⁻ Beam



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After our 200 µs pulse of H- comes out of the ion source, it is accelerated by a series of linear accelerators until it reaches ~ 70 MeV

Acceleration



The drift tubes shield the H^- from the RF resonant wave – so particles only see the RF wave when it accelerates them



ISIS Drift Tube Linac 4

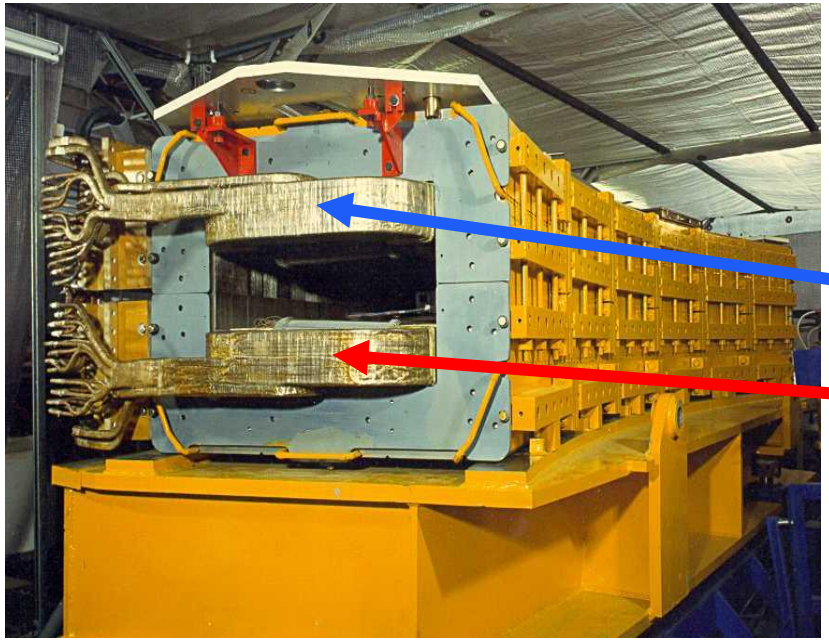


Inside an ISIS Drift Tube Linac

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Magnets: Dipoles bend the beam

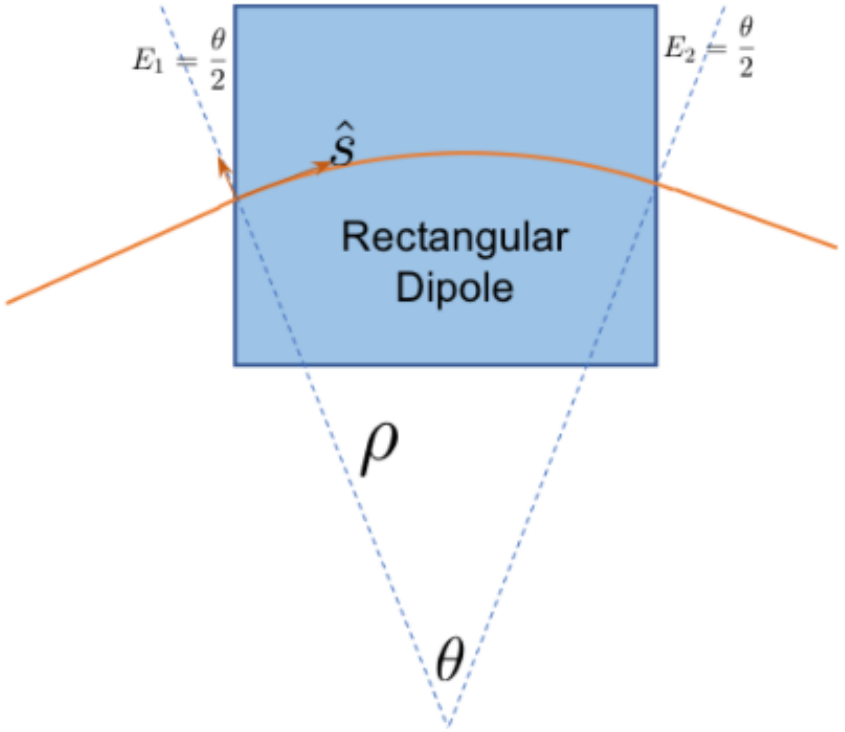


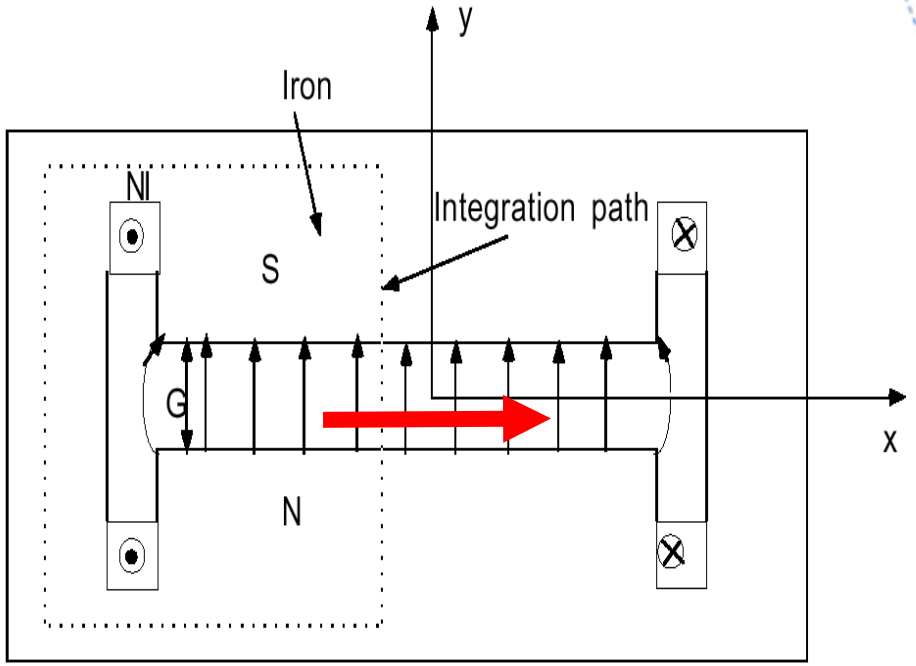
ISIS Main Dipole

Dipoles create a straight magnetic field between 2 poles.

South pole.

North pole.





Charged particles interact with this field and their path is bent. Bending depends on charge/mass and energy of the particle.

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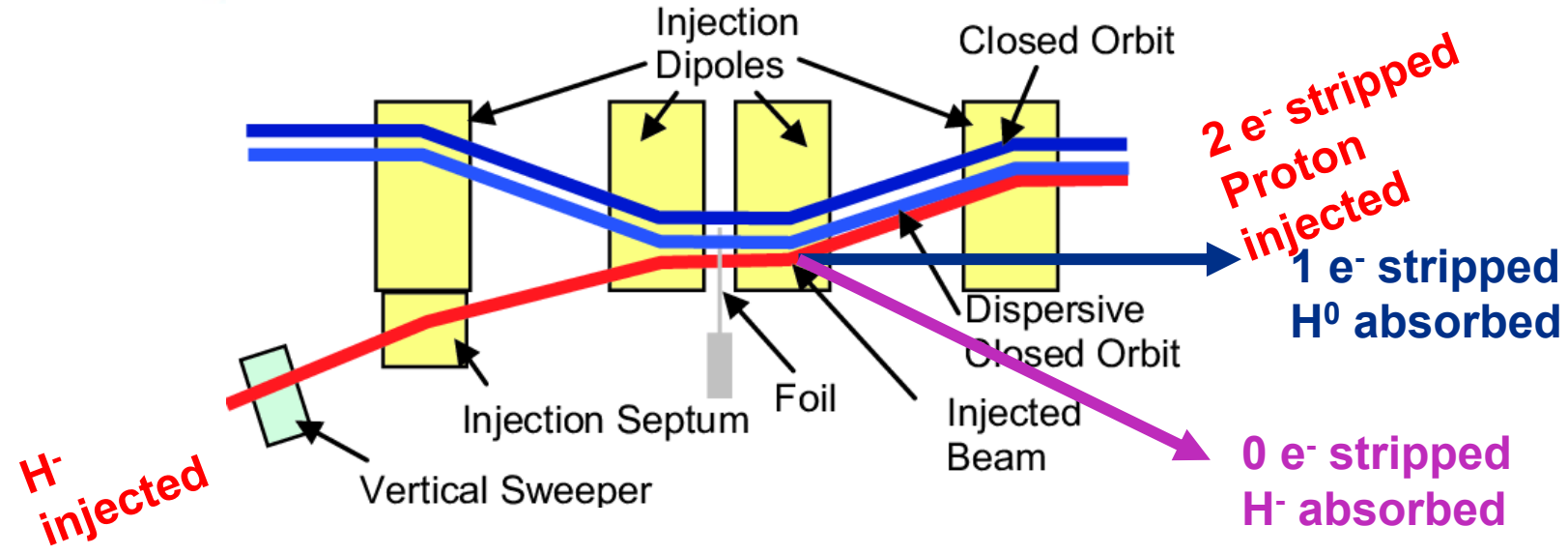
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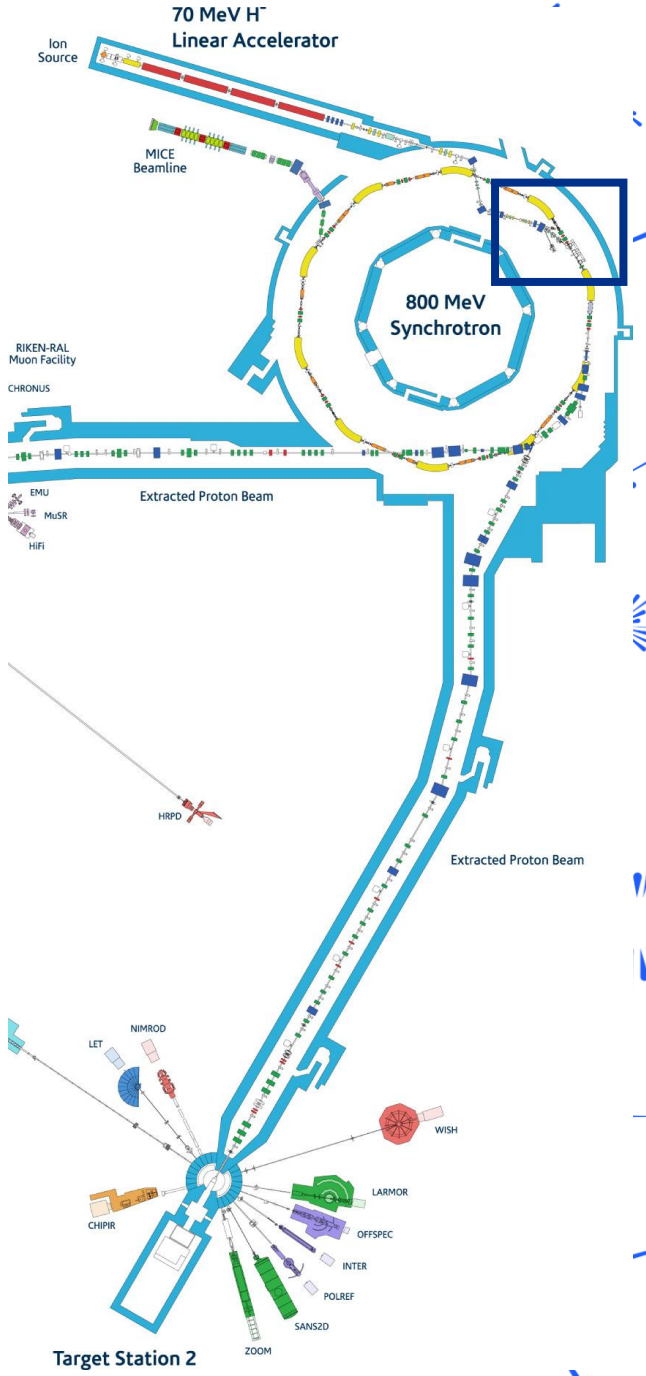
Synchrotron: Injection



Injection into the synchrotron uses foil stripping to remove the electrons from H^- and inject protons into the ring (synchrotron).

We accumulate a few hundred pulses from the ion source / linacs in order to achieve a proton intensity of ~ 3E13. 30 million million protons per pulse.





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ISIS Neutron and Muon Source

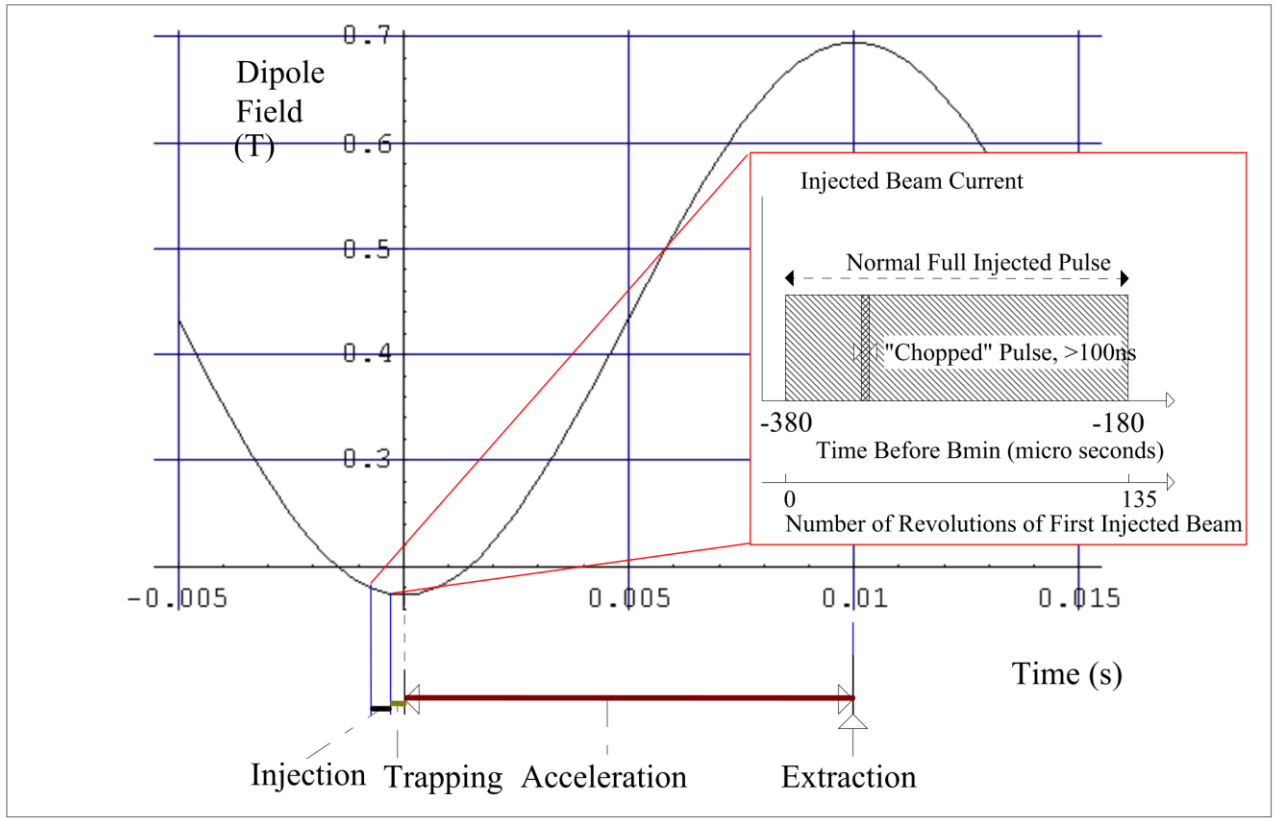


SNS @ ORNL (USA) broke this record in 2008

ISIS Synchrotron: Machine Cycle

- ISIS main magnets (bending and focussing) are powered using AC and DC power. The AC oscillates at 50 Hz.
- 1 second / 50 = 20 milliseconds
- 1 AC oscillation (shown on the right) = 20 ms
- Injection occurs as shown in the diagram, for a few hundred turns (varies depending on need) prior to the lowest point in the AC oscillation.
- The magnetic fields (control bending and focussing of the beam) rise with the AC.
- The RF (controls acceleration) is linked to the magnetic field in the bending magnets, and accelerate accordingly.

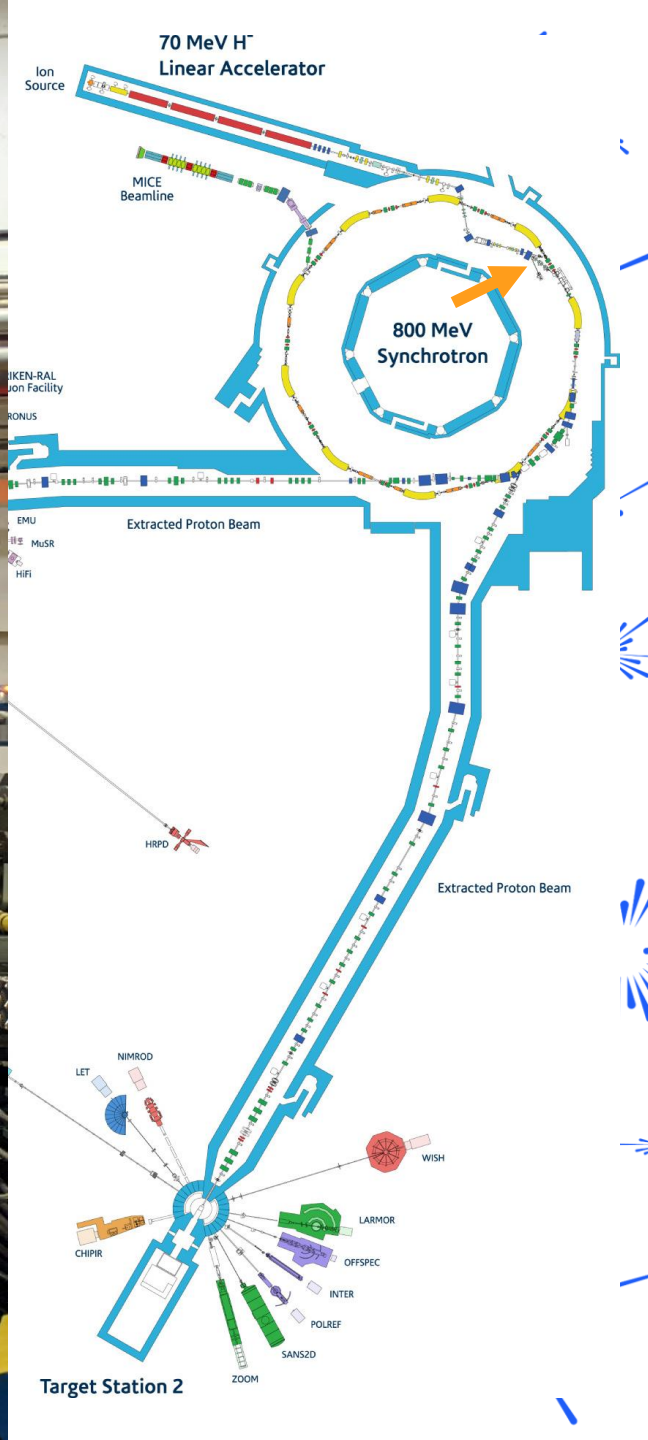
Figure 2.1(b)
Relation of Beam Chopping to Main Magnet Field.



"Special diagnostic methods and beam loss control on high intensity proton synchrotrons and storage rings Circular proton accelerator", C. M. Warsop, PhD Thesis 2002

- Beam is extracted slightly before 10 ms (peak of AC) at 800 MeV (injected at 70 MeV). This equates to an 11.43 times increase in proton energy, taking the protons from 37% to 84% of the speed of light. Protons circulate in the synchrotron for ~ 11,000 turns.

The beam is injected on the inside of the synchrotron





The stripping foil is made from nano-crystalline diamond, and is 1 micron thick.

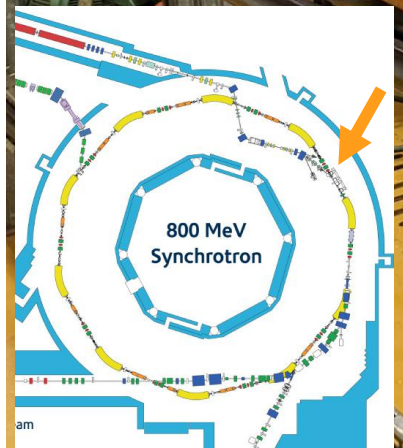
The foil is very delicate and takes a few hours to change (changed every few years currently).

This area is the remote arm that allows us to withdraw the foil from the machine and replace it. The area is very radioactive – hence extra concrete shielding.

We are looking at the outside of the synchrotron, at the position of the injection dipoles and stripping foil.

The 4 boxes (3 white, 1 black) are ion pumps used to maintain vacuum when the injected H- beam hits the stripping foil.

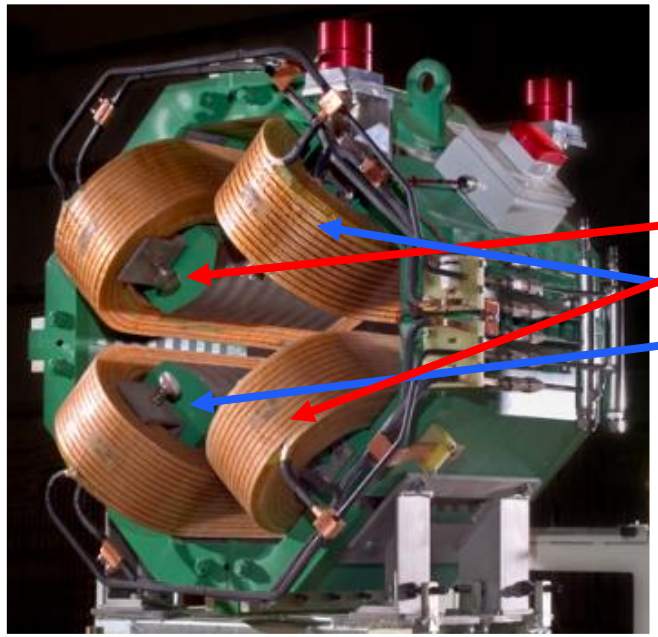
This area is radioactively hot due to the injection stripping process.



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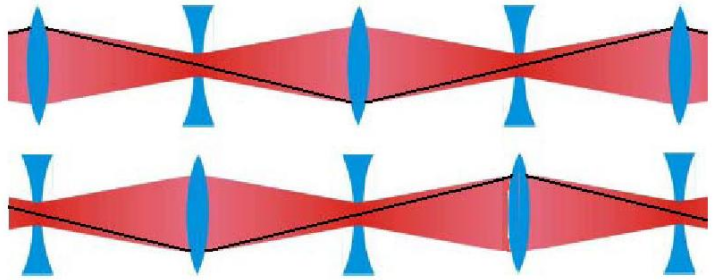
Magnets: Quadrupoles focus the beam



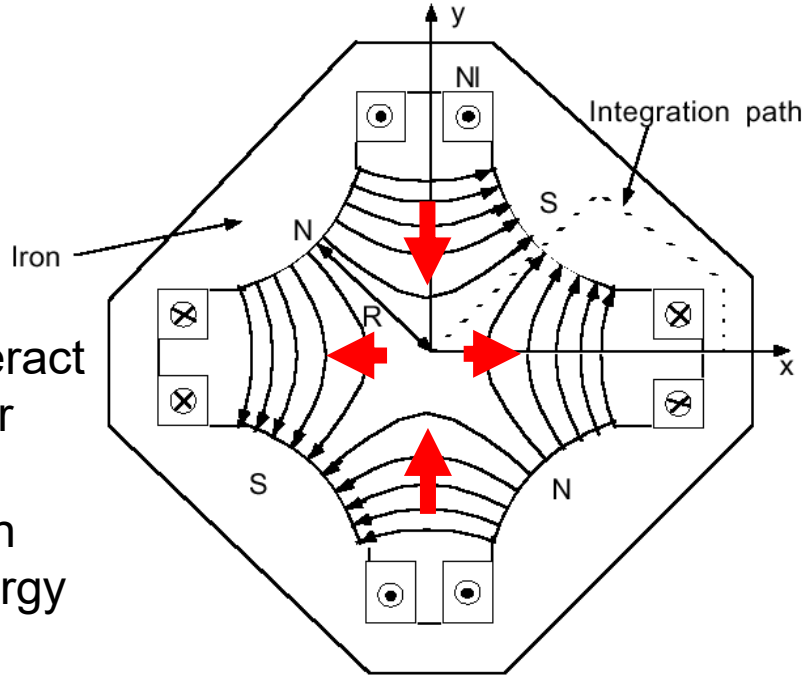
Quadrupoles create a curved magnetic field between 4 poles.

North poles.

South poles.



Beam Trajectory



Magnetic Fields /Forces

Charged particles interact with this field and their path is deflected. Deflection depends on charge/mass and energy of the particle, and its position within the field.

A particle travelling through the centre of the field will feel no force.

Quadrupoles focus in one transverse plane (here vertical), and defocus in the other transverse plane (here horizontal).

We use quadrupoles in doublets (rotated by 90 degrees) to give a net focussing force.

Main Quadrupole

Main Quadrupole

Main Dipole

Trim Quadrupole



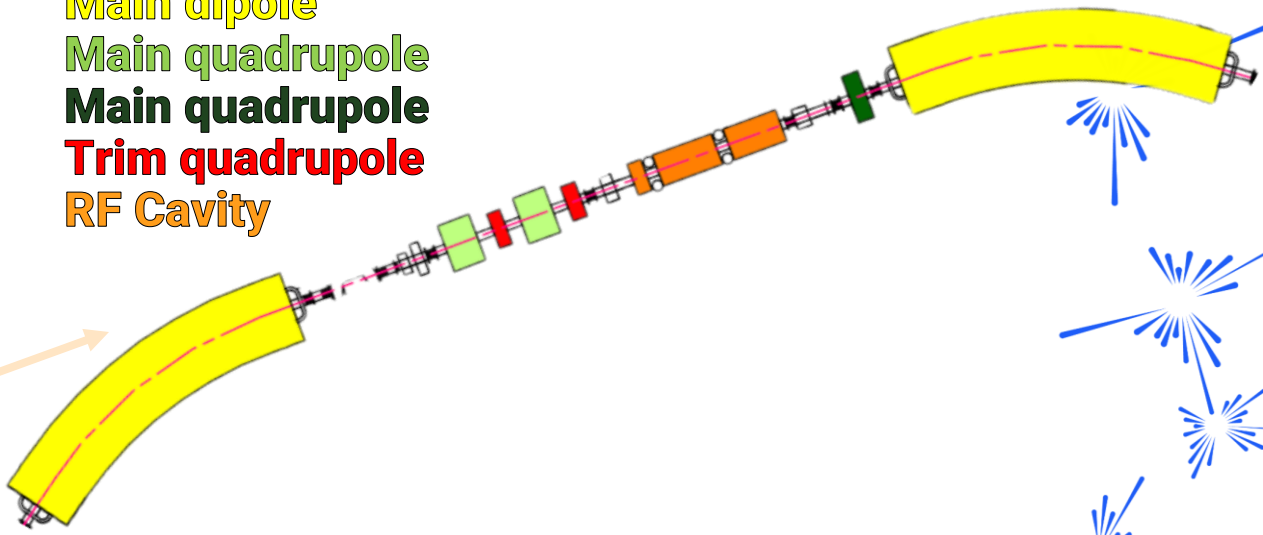
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Beam Focussing

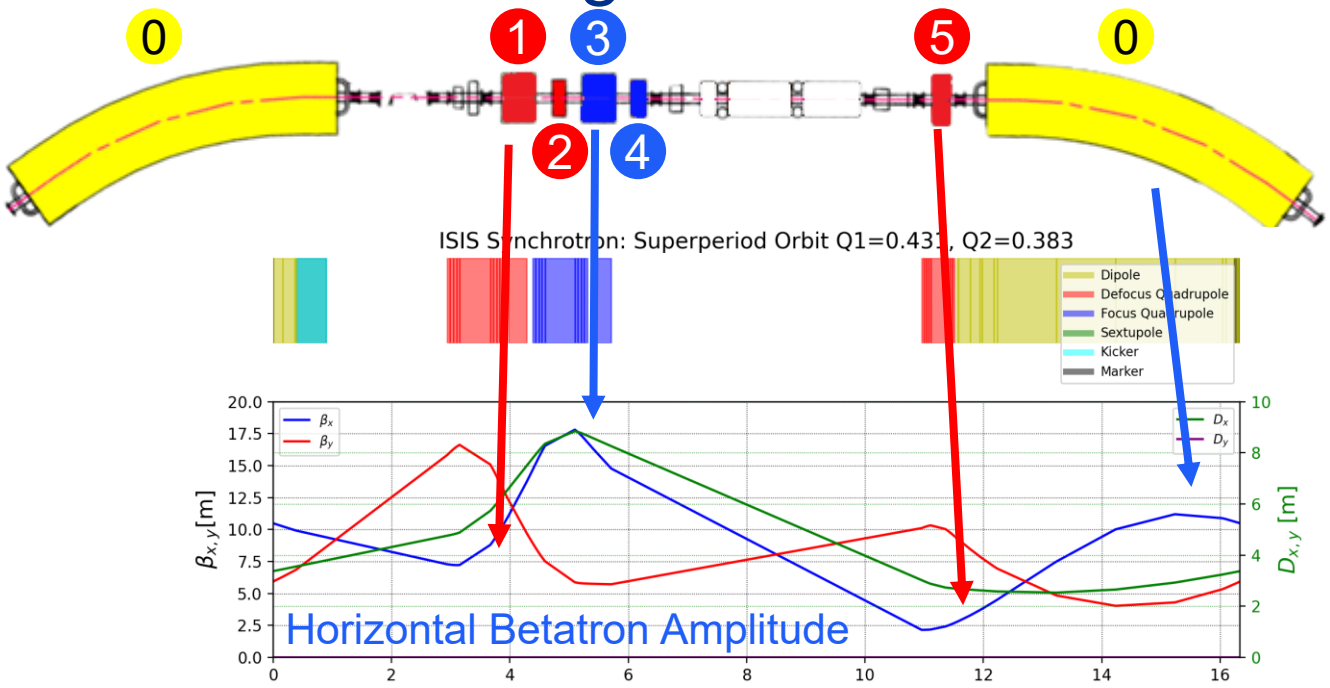


Main dipole
Main quadrupole
Main quadrupole
Trim quadrupole
RF Cavity



Let us consider a single super-period of the ISIS synchrotron (including only one main dipole)

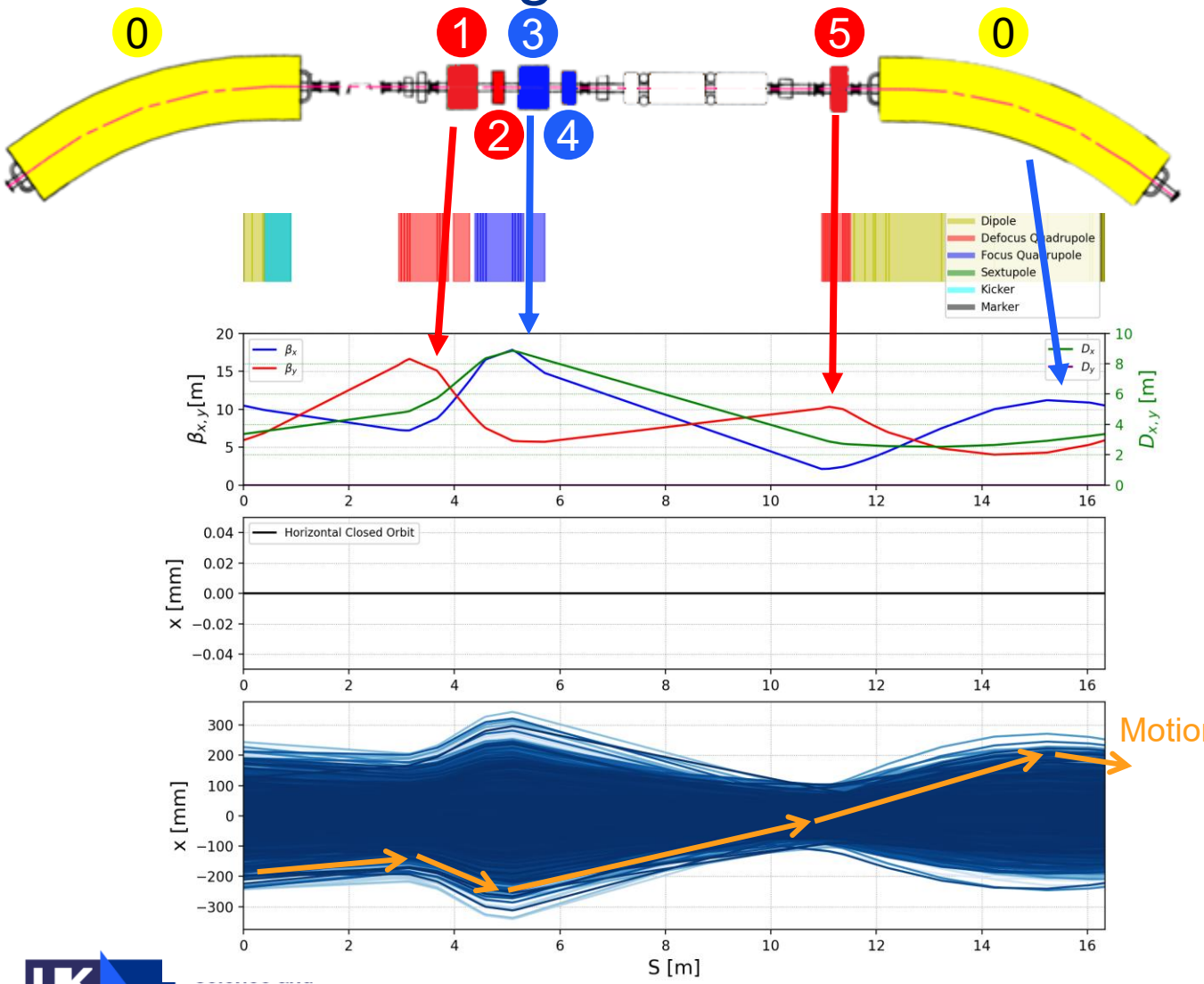
Beam Focussing



0. Main dipole
1. **Main Defocussing quadrupole**
2. **Trim Defocussing quadrupole**
3. **Main Focussing quadrupole**
4. **Trim Focussing quadrupole**
5. **Main Defocussing quadrupole**

β_x, β_y = horizontal / vertical betatron amplitude
Parameterisation of beam size due to quadrupole fields (focussing lenses)

Beam Focussing



0. Main dipole
1. **Main Defocussing quadrupole**
2. **Trim Defocussing quadrupole**
3. **Main Focussing quadrupole**
4. **Trim Focussing quadrupole**
5. **Main Defocussing quadrupole**

β_x, β_y = horizontal / vertical betatron amplitude
Parameterisation of beam size due to quadrupole fields (focussing lenses)

Motion of a single particle

Horizontally: Beam envelope follows the shape of β_x

This shows us that the beam is oscillating in width through the super-period

What is the Tune?

Tune = Q = number of betatron oscillations a particle makes in a single orbit.

In a bunch, each particle has an individual tune. A **bunch** of particles (outside the scope of this talk) **has a tune-spread.**

Tune (or working point Q_x, Q_y) is usually **monitored in transverse planes,** and **is controlled by varying focussing fields** (trim quadrupoles)**.**

We control the working point, particles have a known spread around this point. Cause of spread results in damping (reduction) of particle tune only.

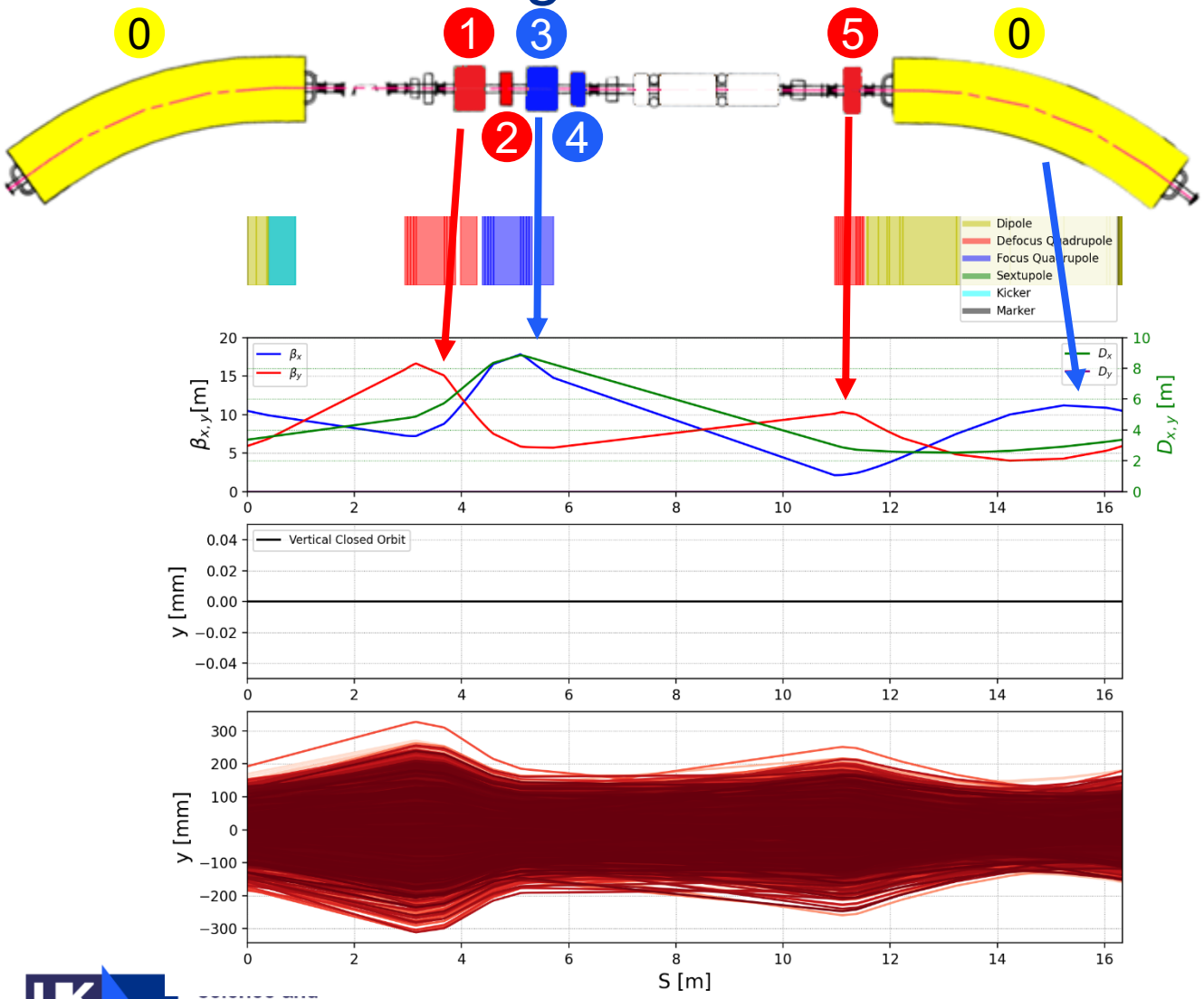
Example ISIS working point (Q_x, Q_y) = **(4.31, 3.83)**

Operational Goals:

Predict and **visualise** the working point of the synchrotron at a specific time in the acceleration cycle (at a specific energy)

Identify discrepancy between prediction and measurements

Beam Focussing



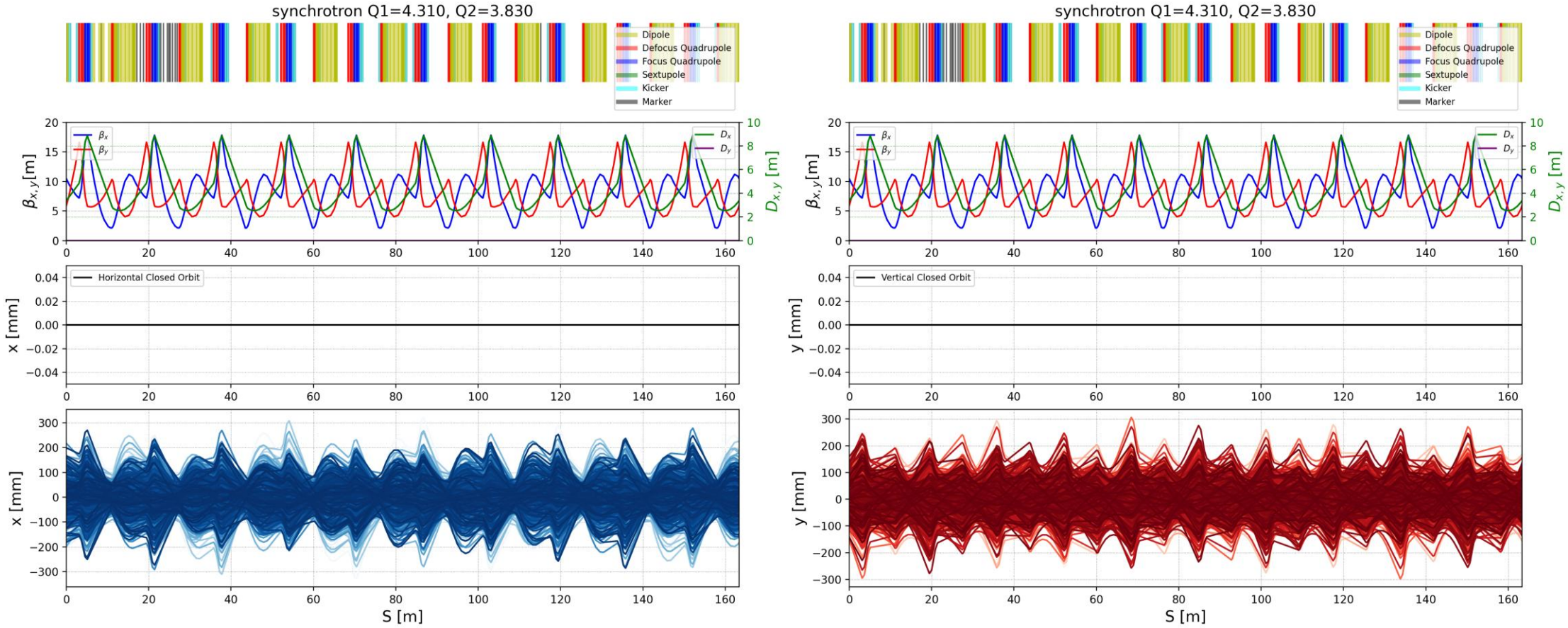
0. Main dipole
1. **Main Defocussing quadrupole**
2. **Trim Defocussing quadrupole**
3. **Main Focussing quadrupole**
4. **Trim Focussing quadrupole**
5. **Main Defocussing quadrupole**

β_x, β_y = horizontal / vertical betatron amplitude
Parameterisation of beam size due to quadrupole fields (focussing lenses)

Vertically: Beam envelope follows the shape of β_y

This shows us that the beam is oscillating in height through the super-period

Beam Focussing

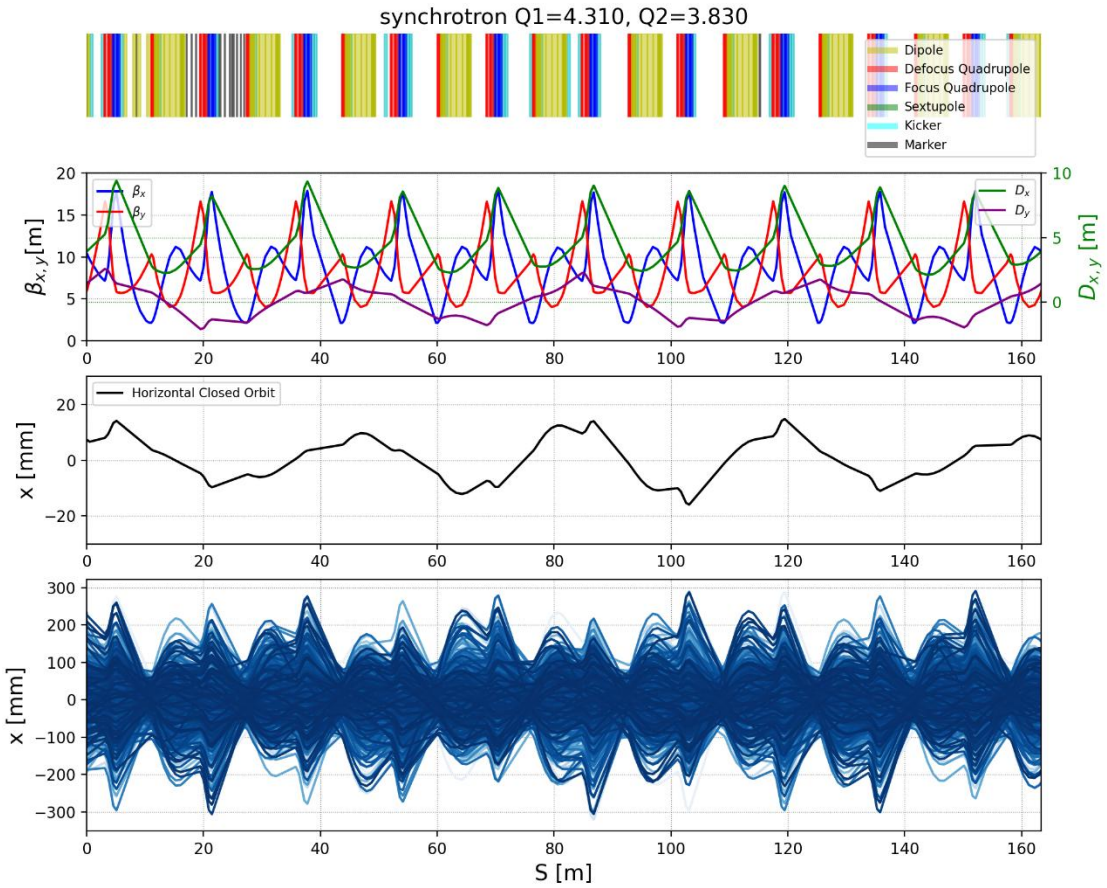


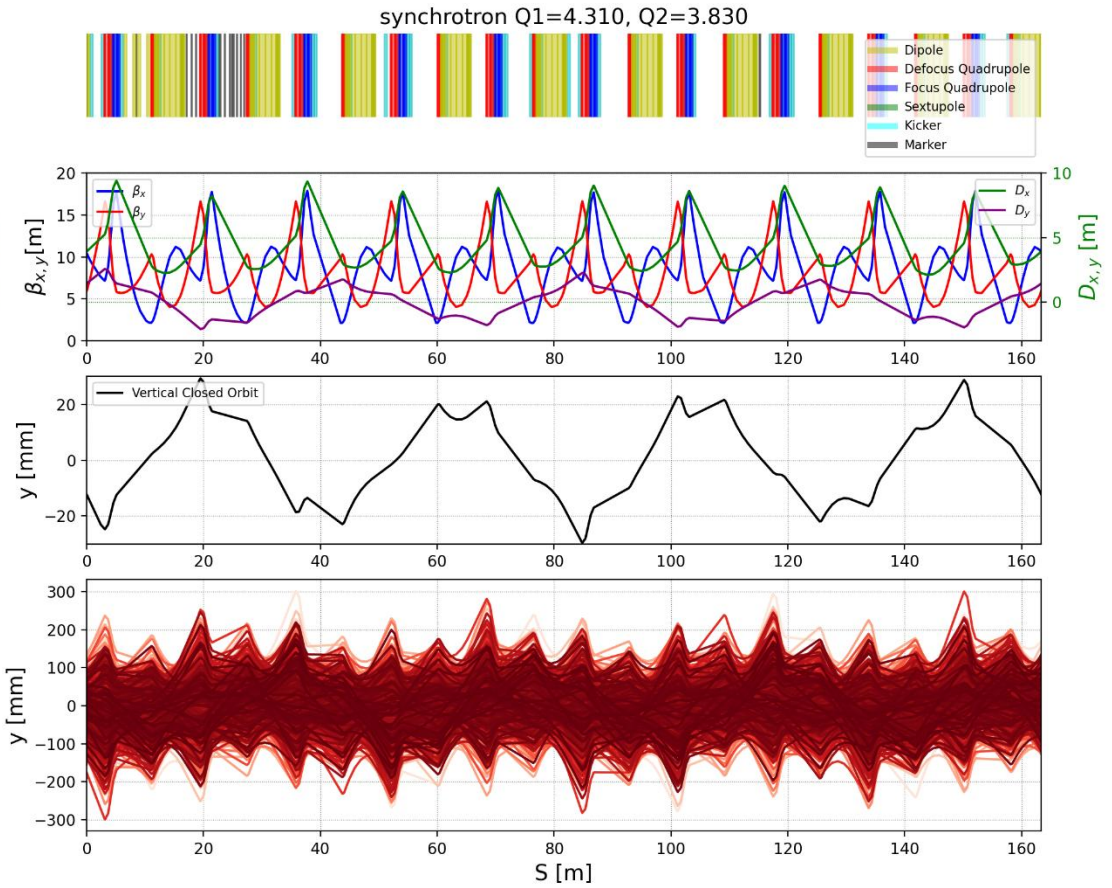
As the beam travels through the 163 metres of synchrotron, it performs oscillations in x, y, and z (not shown).
We may describe this motion using partial differential equations, and simple harmonic motion (outside the scope of our work experience).

Closed Orbit

Add ~2 mrad dipole kicks in H and V
R0HD1 +30A, R0VD1 +55A equivalent @ 0 ms

Closed Orbit (mean beam path) for reference particle is **perturbed.** Gives ≈± 20 mm oscillation.





What is the Closed Orbit?

Measured Closed Orbit = the average path taken by a beam of charged particles in an accelerator at a specific time in the acceleration cycle

Simulated Closed Orbit = Design trajectory

Context	*Description*	*Key Assumptions*
Ideal Closed Orbit	**Design trajectory** in a perfect lattice	No errors, design energy
Distorted Closed Orbit	Actual periodic orbit with **imperfections**	Includes misalignments and corrections
Centroid (measured) Orbit	Average beam path	Includes collective effects
Off-Momentum Closed Orbit	Path of particle with $\delta \neq 0$	Defines dispersion
Turn-by-Turn Orbit	Periodic orbit over many turns	Useful in diagnostics
Bare Orbit	Orbit from survey errors only	Excludes correctors
Matched Envelope	Equivalent in linac or transfer line	Applies in linear transport

Operational Goals:

Predict and **visualise** the average trajectory of the beam at a specific time in the acceleration cycle (at a specific energy) using the design trajectory plus the corrector dipole settings

Use this as a basis of a **controls tool to correct** the measured average beam path

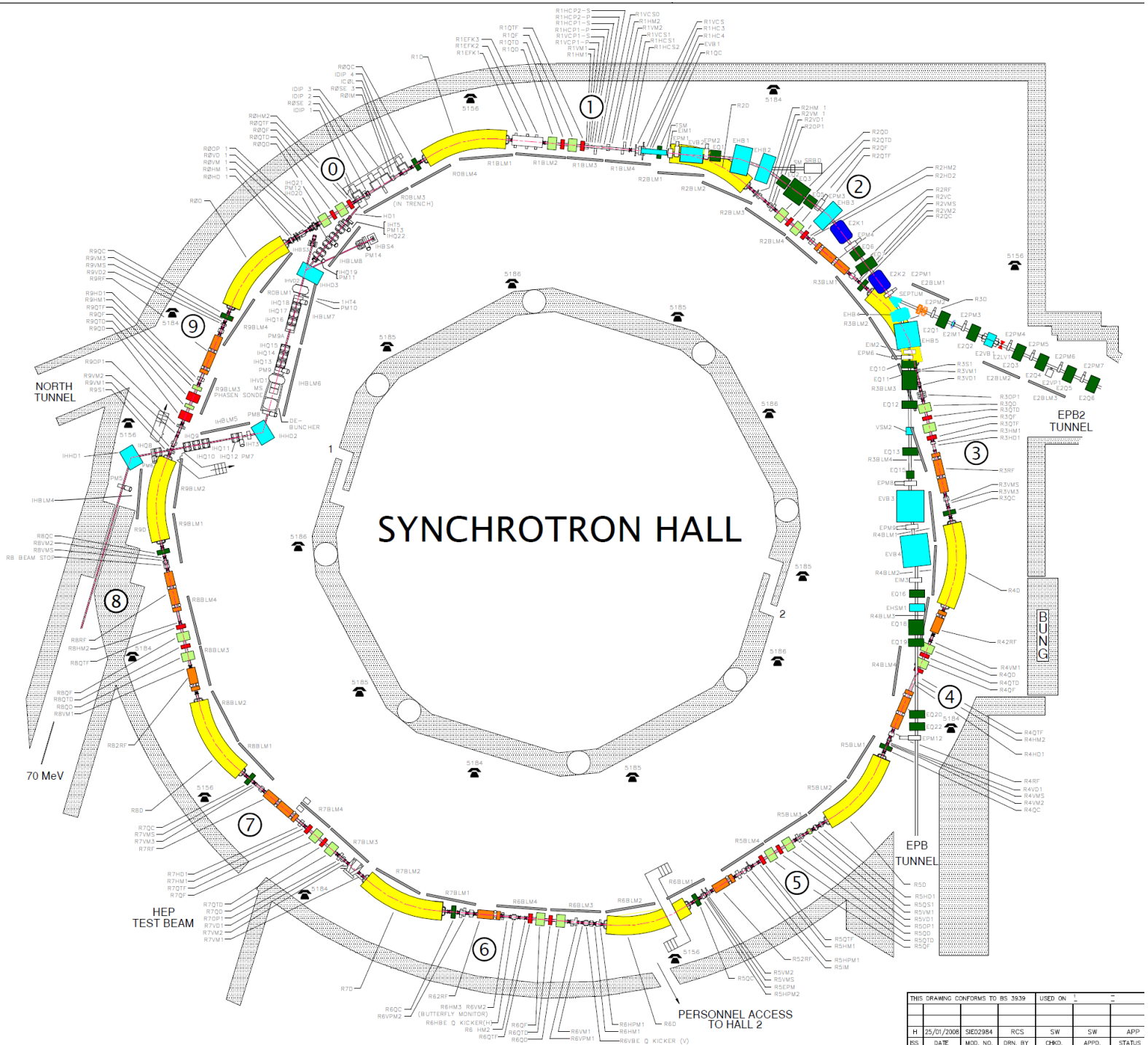
Use this as the basis of a **virtual diagnostic**: envelope vs aperture **loss prediction**

Identify **discrepancy** between prediction and measurements as a basis of **fault detection**

The ISIS Synchrotron consists of 10 identical (in terms of magnet layout) super-periods.

Numbered here from 0 – 9.

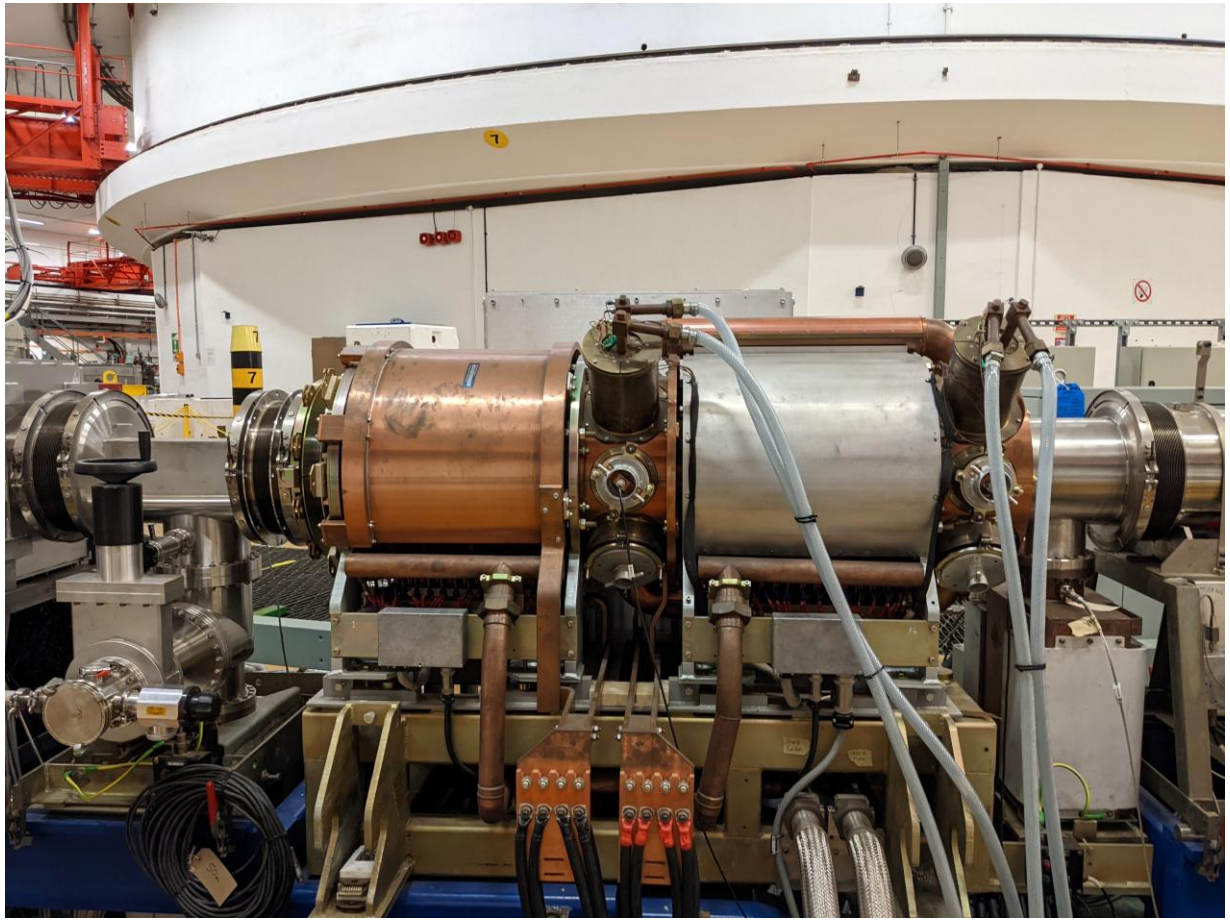
Unfortunately when the machine is on radiation is too high for entry – thus your tours at 14:00 will be of target station 2.

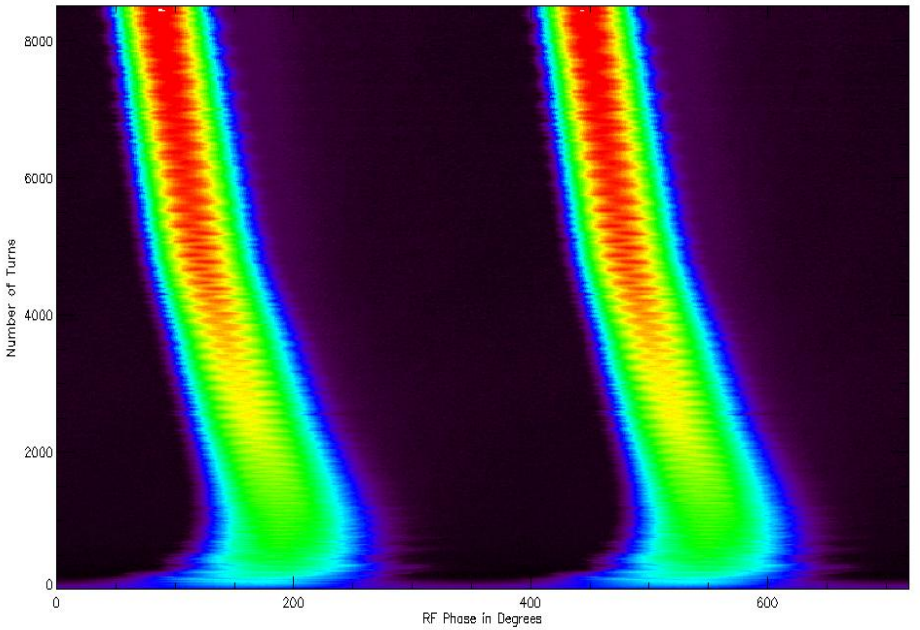


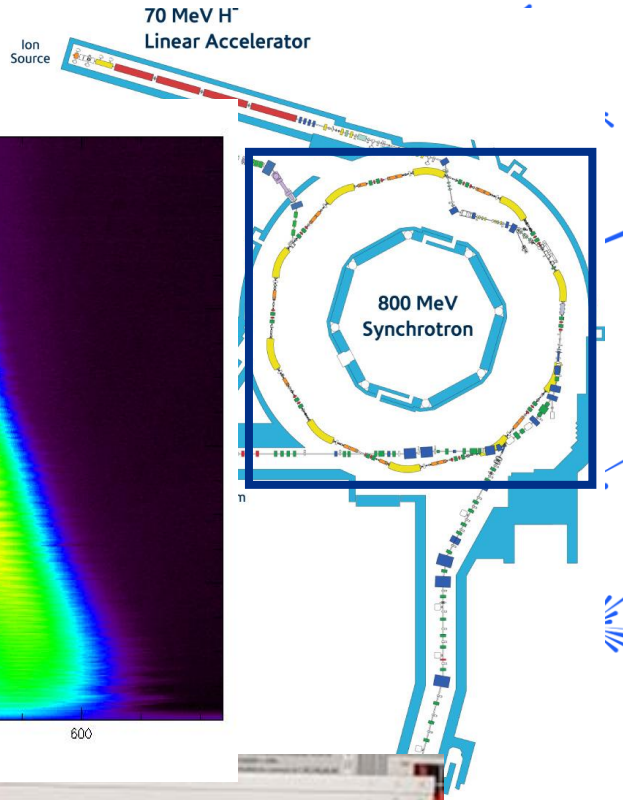
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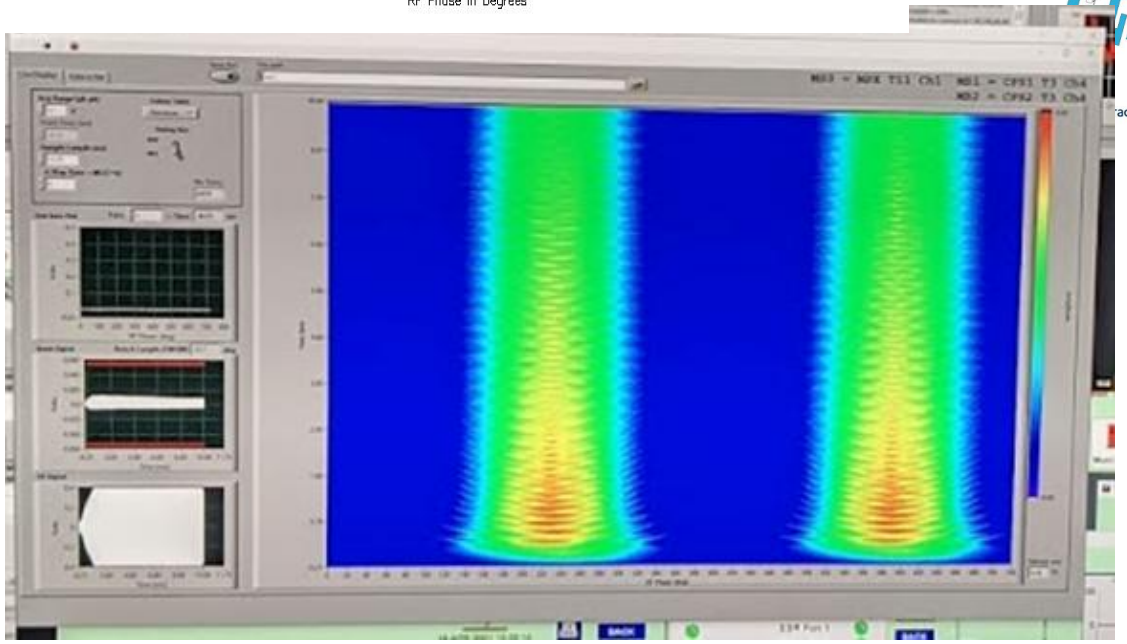
ISIS Neutron and Muon Source

Synchrotron: Longitudinal Gymnastics







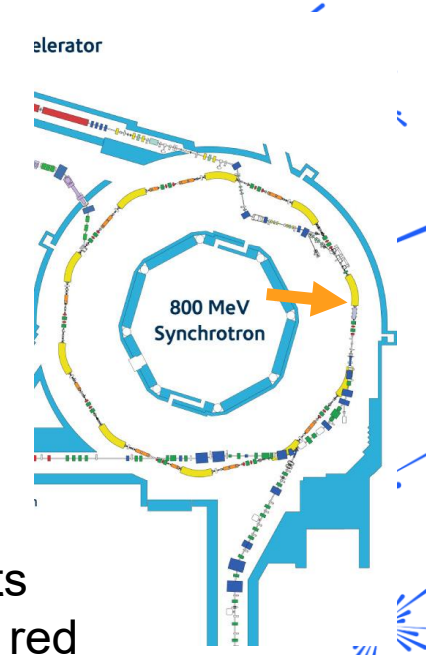


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Synchrotron: Extraction





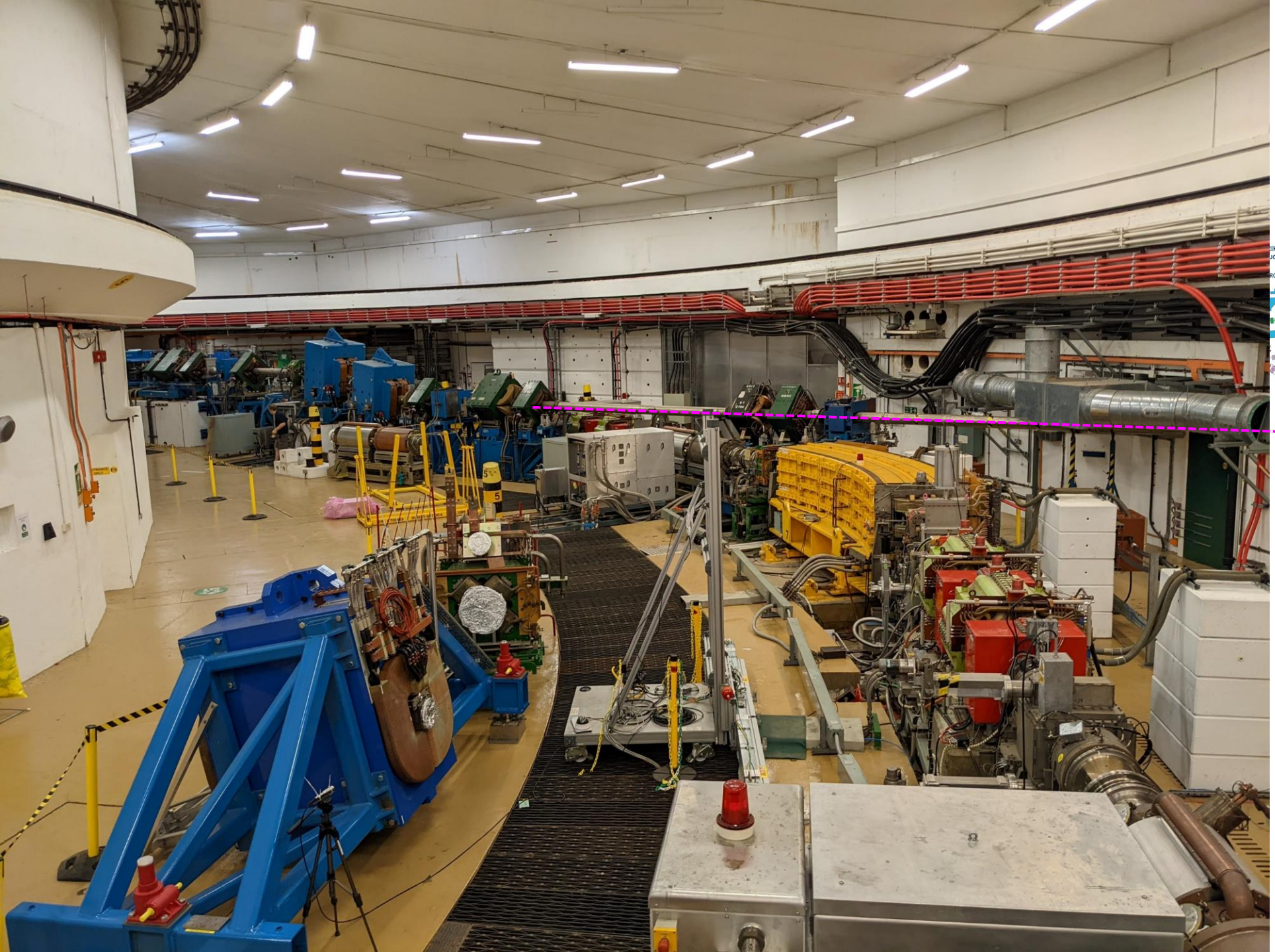
Extract kicker magnets (powered by massive red cables) fire rapidly with a 150 ns rise time at the end of the acceleration cycle.

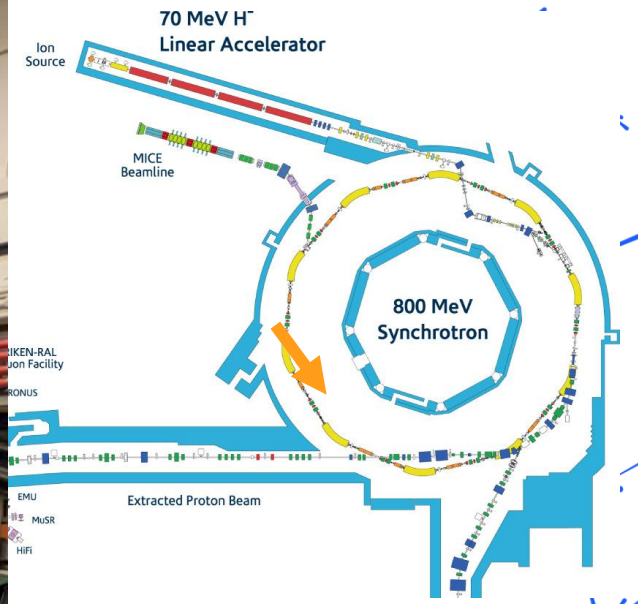
The kickers give the beam a vertical kick of a few mm so it can be extracted and sent to the target stations to produce neutrons and muons.



Extracted beam kicked upwards by extract kickers sees the blue dipole field and is bent upwards out of the synchrotron.

Circulating beam travelling around the synchrotron for ~11,000 turns before the extract kickers fire sees no magnetic field.





To TS1

After being extracted the beam is transported above the synchrotron (super-periods 2/3/4) and sent to 2 target stations



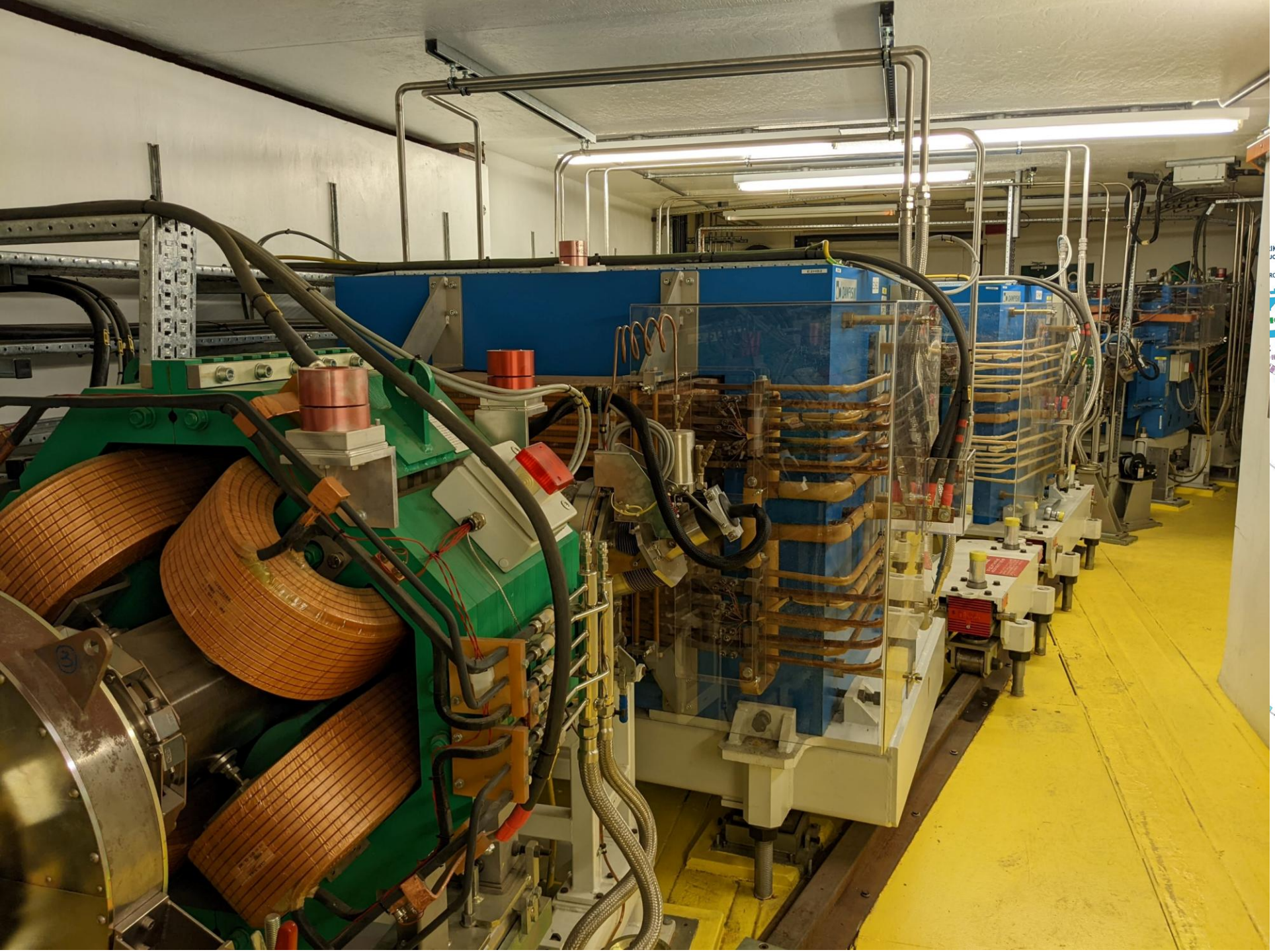


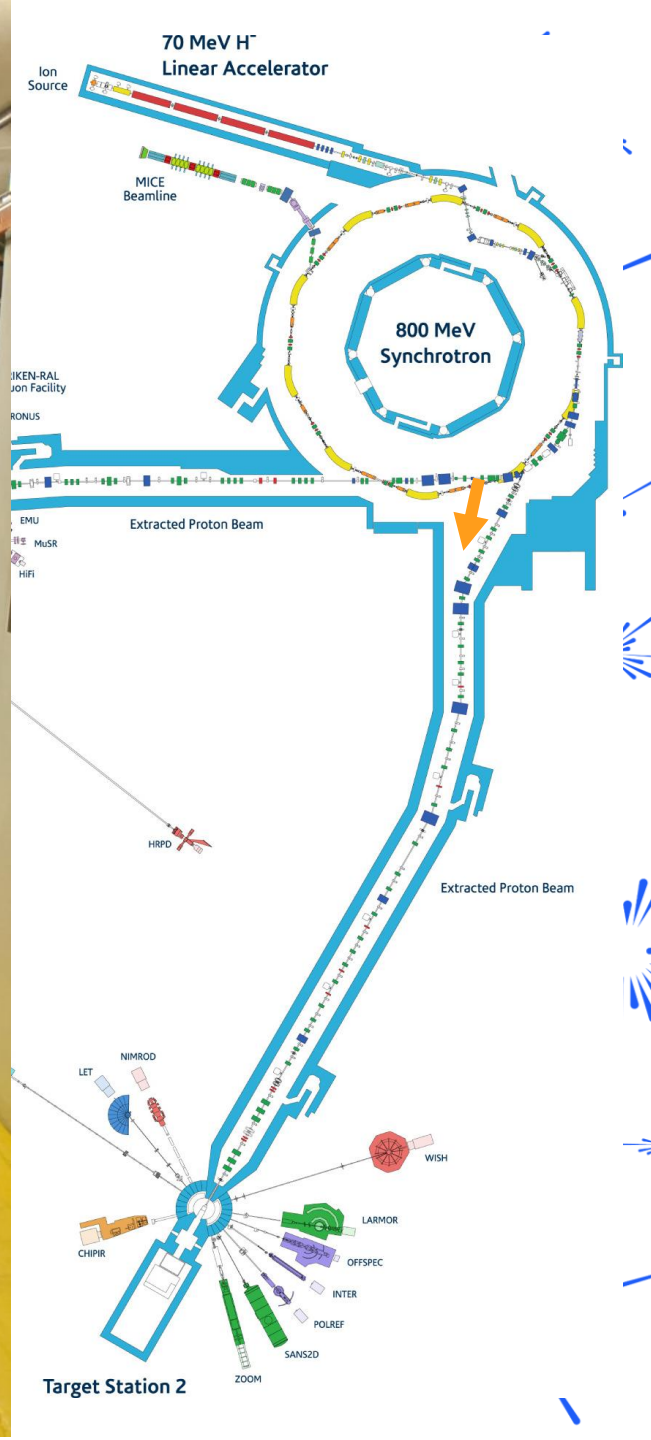


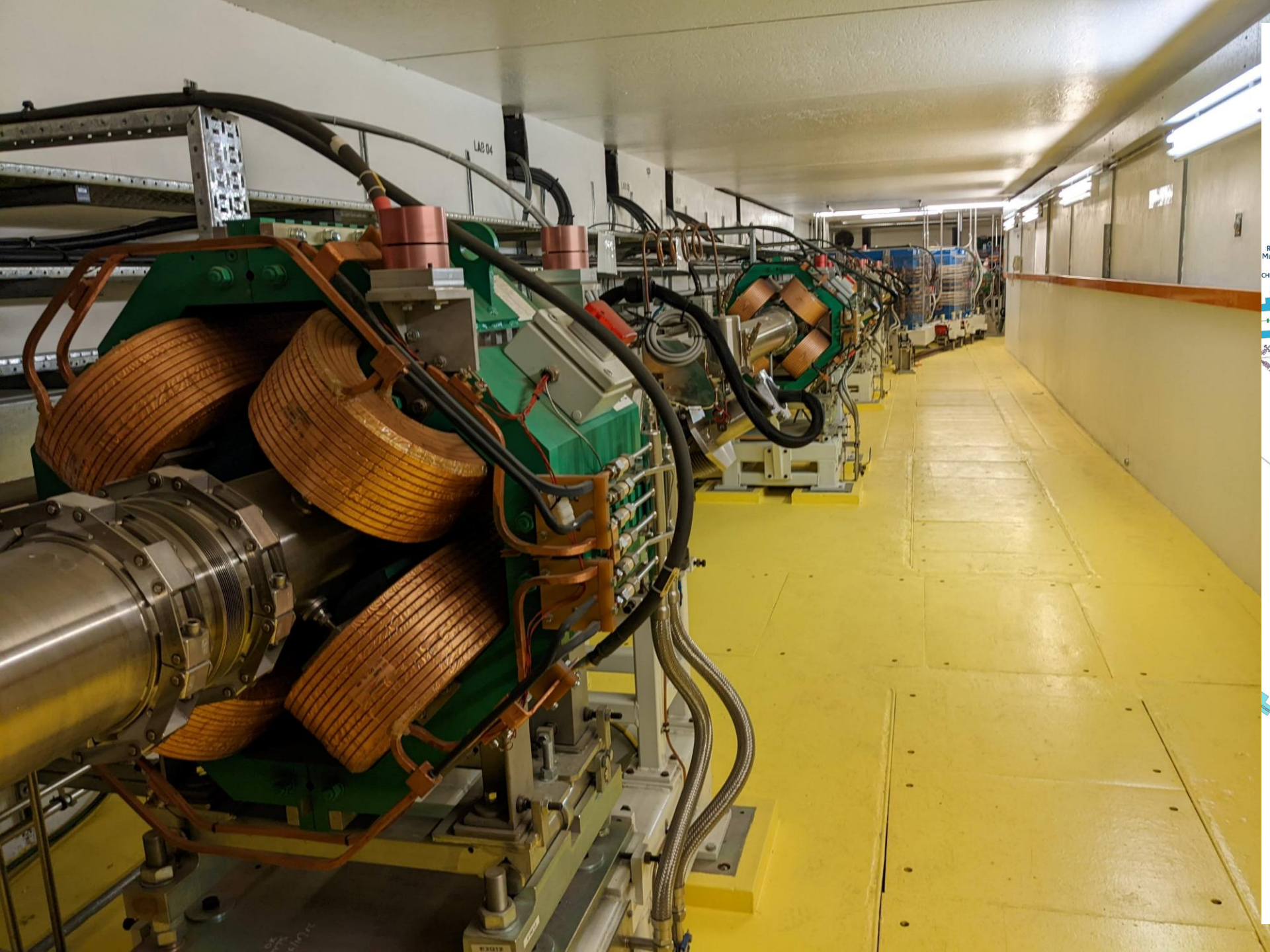
Target station 2
extraction line

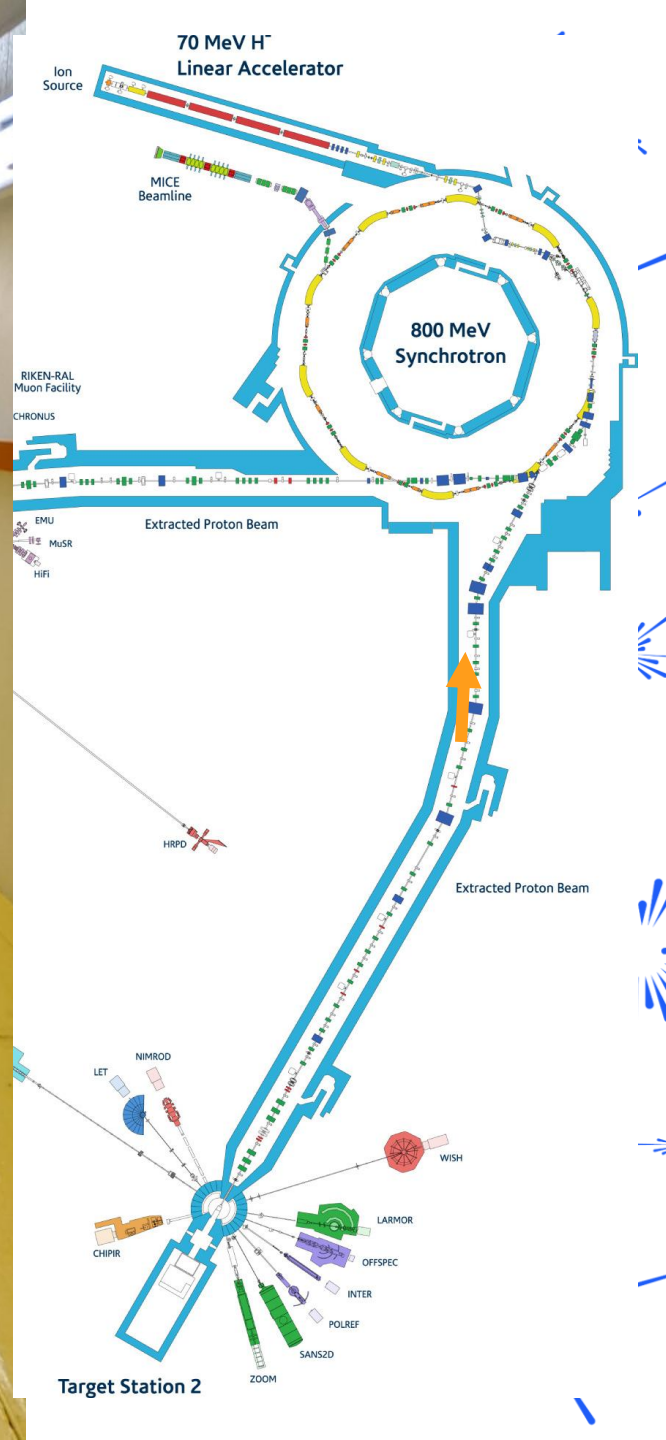
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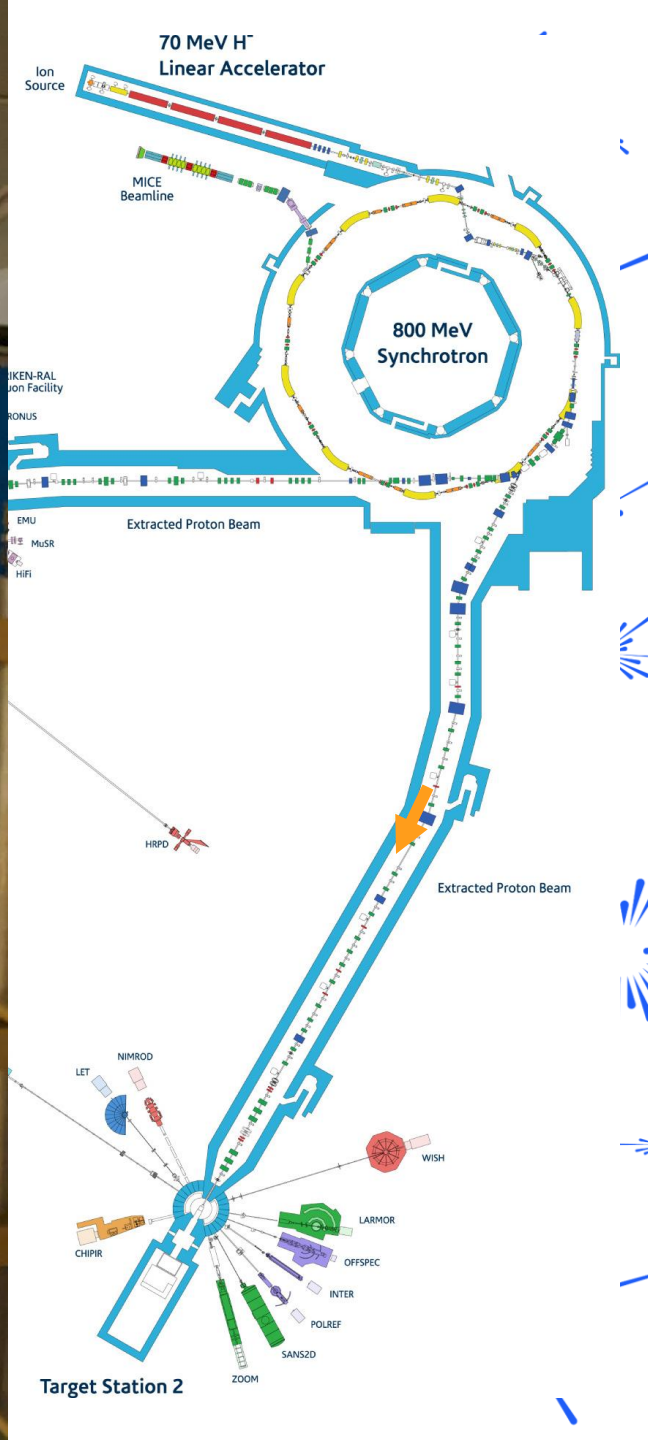



70 MeV H⁻
Linear Accelerator
Ion Source
MICE Beamline
800 MeV Synchrotron
Extracted Proton Beam
EMU
MuSR
HiFi
HRPD
Extracted Proton Beam
NIMROD
LET
WISH
LARMOR
CHIPIR
OFFSPEC
INTER
POLREF
SANS2D
ZOOM
Target Station 2



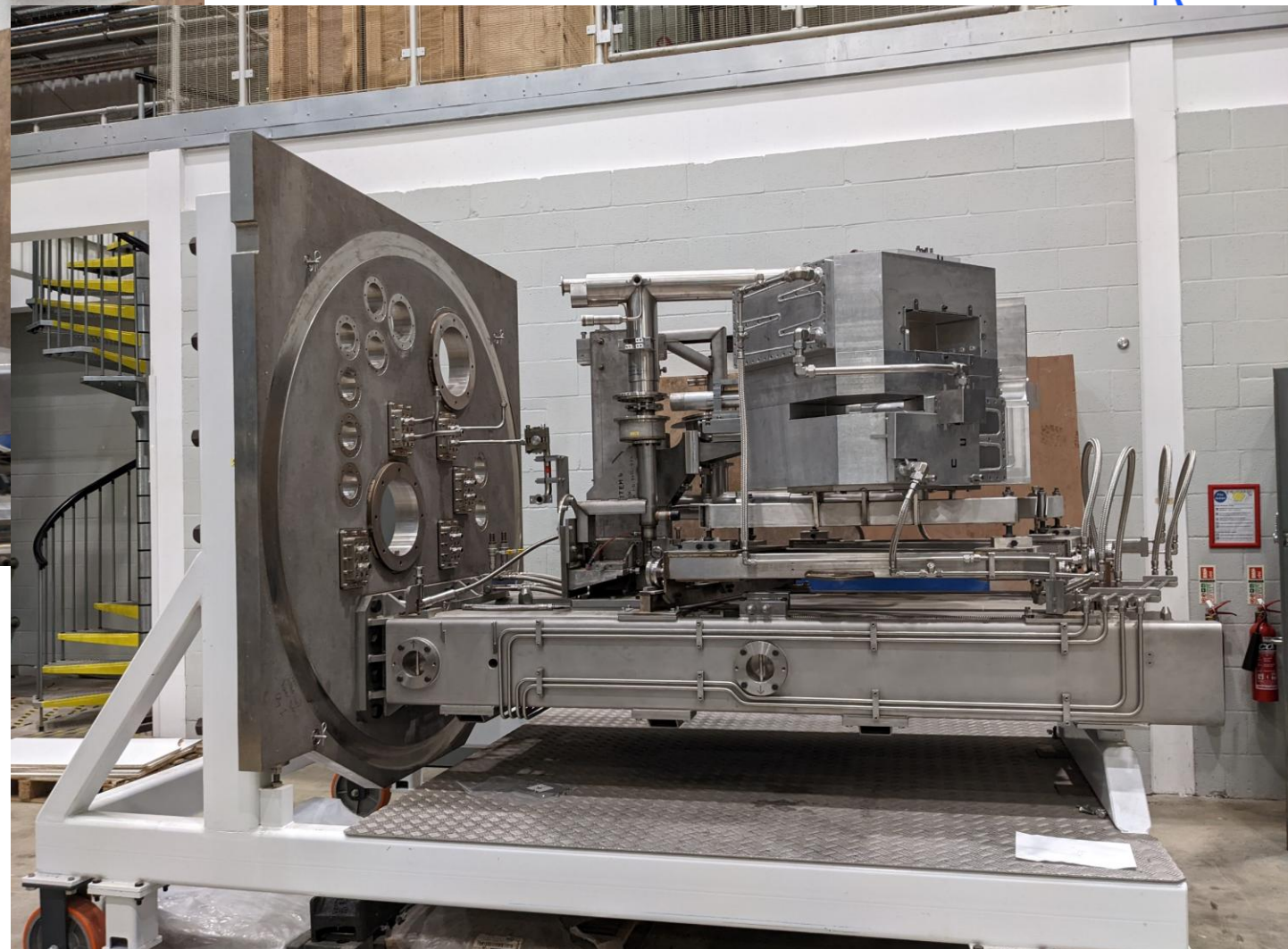

70 MeV H⁻
Linear Accelerator
Ion Source
MICE Beamline
800 MeV Synchrotron
RIKEN-RAL Muon Facility
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70 MeV H⁻
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LARMOR
CHIPIR
OFFSPEC
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SANS2D
ZOOM
Target Station 2

Target Station 2 Mock Target





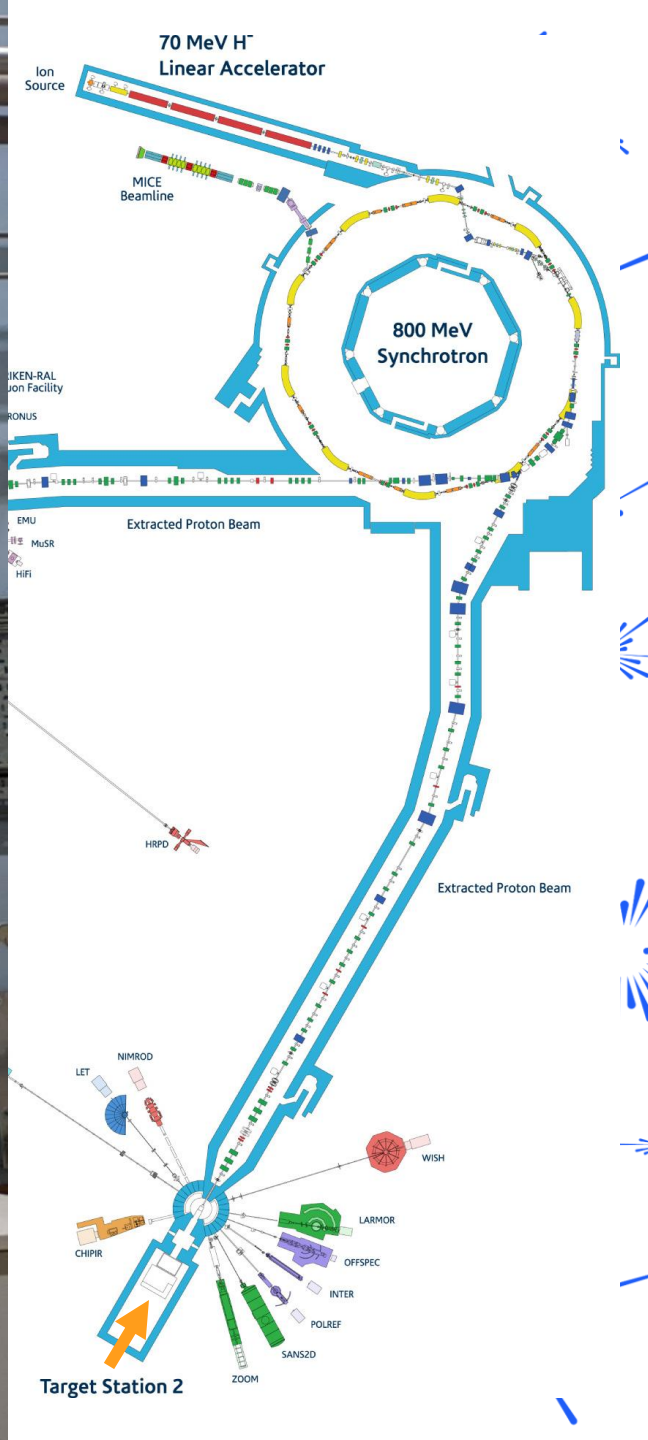
Solid Tungsten target (central cylinder) is hit with 800 MeV proton beam. Produces neutrons via spallation. Neutron flux doubled by beryllium reflector. Neutron energy spectra controlled by moderator liquids.

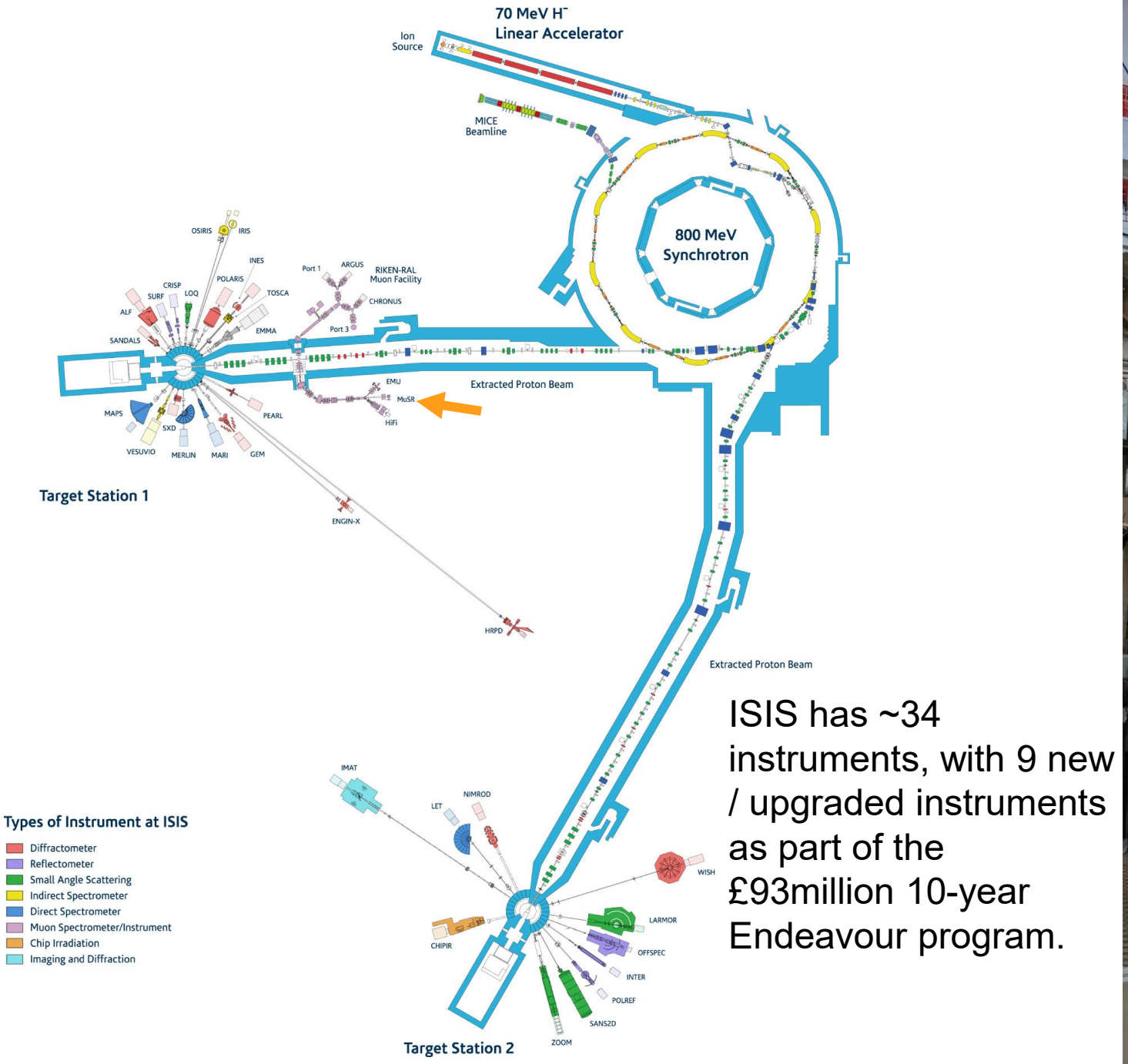
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The targets require a lot of hardware – they are very radioactive and thus maintenance is performed with remote manipulation behind lead glass.





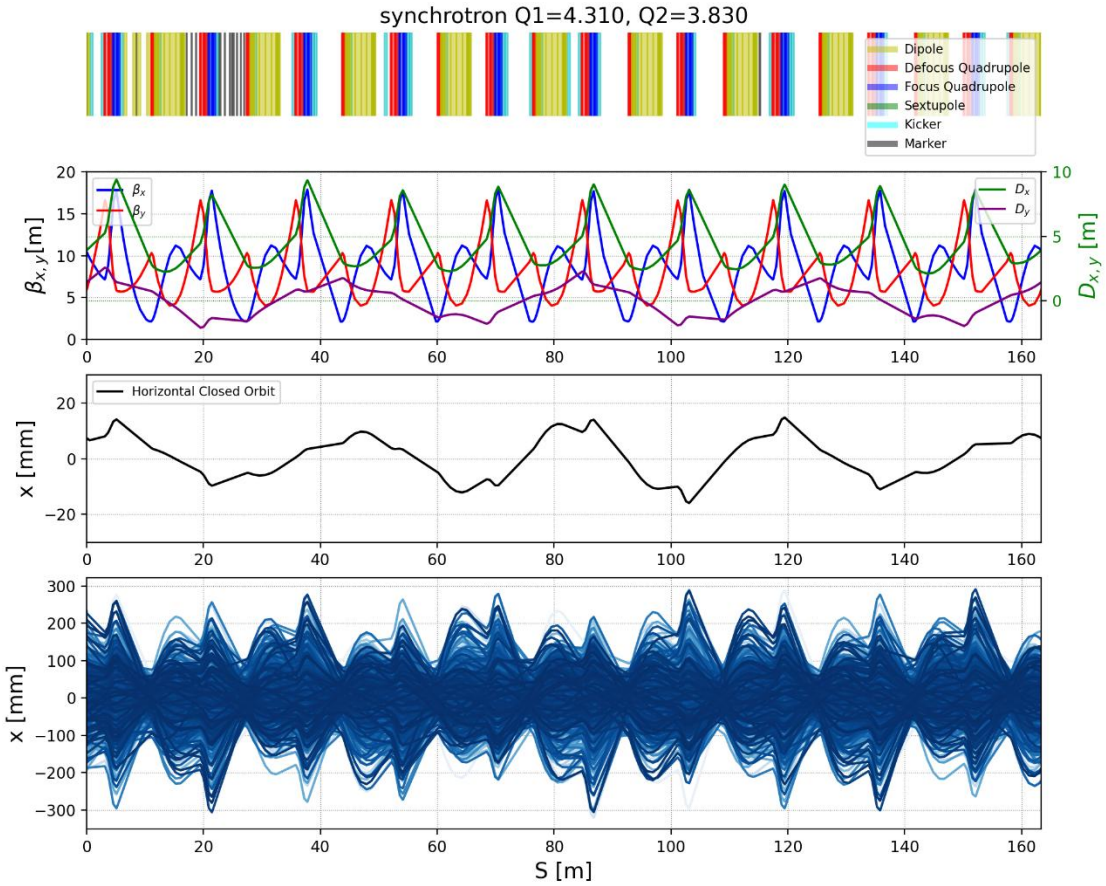
ISIS has ~34 instruments, with 9 new / upgraded instruments as part of the £93million 10-year Endeavour program.



https://www.isis.stfc.ac.uk/Pages/Science.aspx

Quiz – test your understanding

- What does a dipole magnet do to a charged particle beam?
- What does a quadrupole magnet do to a charged particle beam?
- What is the closed orbit?
- What is the tune?
- How do we change (correct) the closed orbit?
- How do we change the tune?



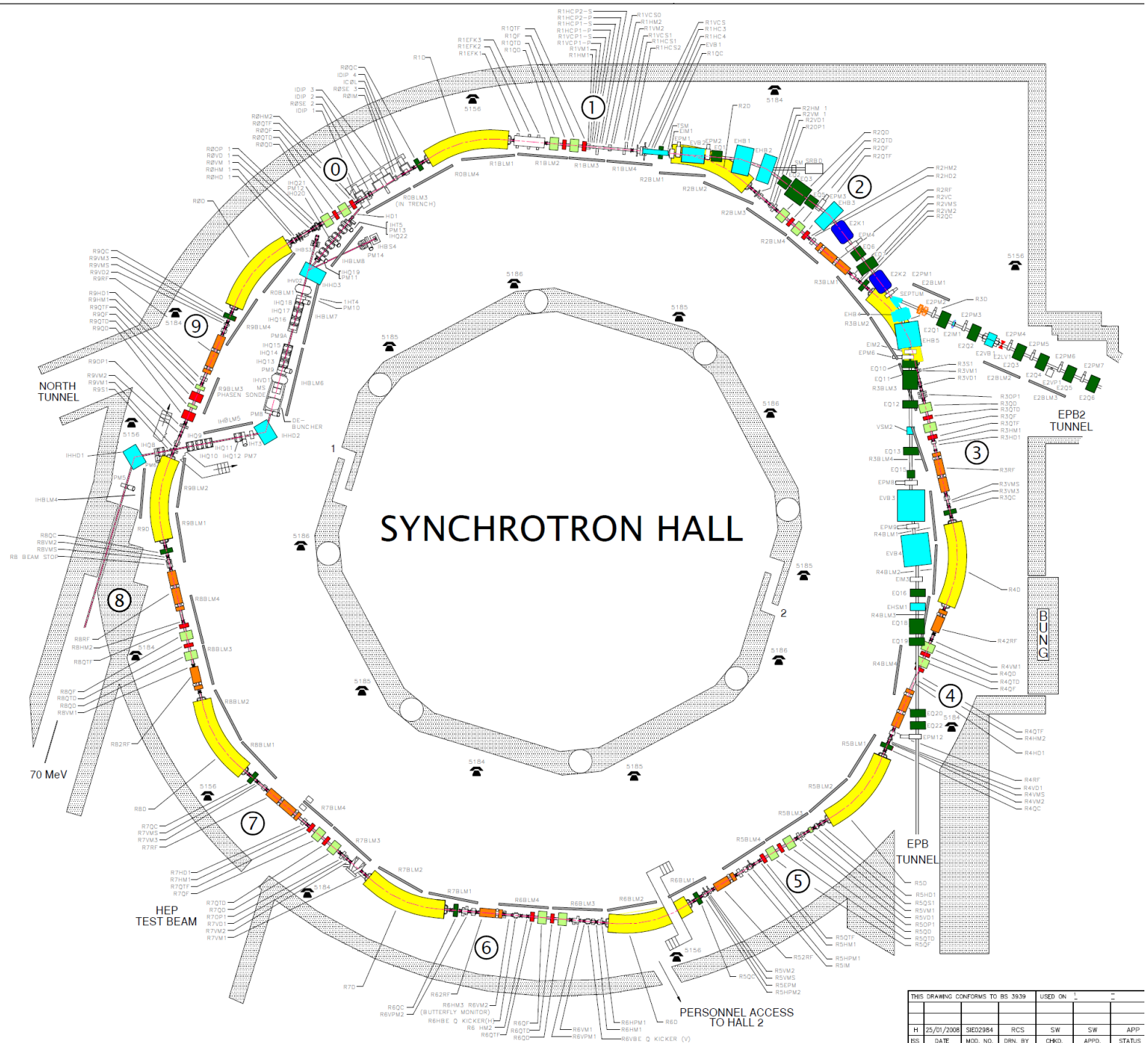
Work Experience Aims

What we will be working on



The story so far

- 2022:
  - (Online 3 x 1 week x 5) Students made **python widgets** to **plot particle distributions** and **beta** functions, **track particles**, and **plot** their **tracks**
- 2023:
  - (26 x 2 weeks) Students communicated with ISIS diagnostics via **MQTT** retrieving beam loss signals and made several **pyQT GUIs** to **visualise beam loss** in the synchrotron
- 2024:
  - (12 x 1 week) Students communicated with **EPICS** and **plotted orbits and tunes** throughout the 10 ms machine cycle, some performed **machine learning** tasks
- 2025:
  - (12 x 1 week) Students produced Streamlit GUIs to plot orbits, tunes, envelopes using a lightweight model backend
  - These GUIs proved the effectiveness of Streamlit as a python accessible GUI development tool
- **2026**
  - **(12 x 1 week)** You will produce **Streamlit GUIs** aided by **generative AI**, building on **a robust and** (hopefully) **complete ISIS synchrotron physics model backend.**
  - **Your GUIs will be the prototype of operational optics prediction tools, to be deployed in the controls system this (financial) year.**



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Get live and archived settings from EPICS controls system

Plot synchrotron beam orbit in Streamlit GUI using EPICS settings

Plot synchrotron beam focussing (tune) in Streamlit GUI using EPICS settings

Plot synchrotron beam envelope (orbit+tune) in Streamlit GUI using EPICS settings

Extension: Correct orbit from error table or measurement

Extension: Compare envelope with aperture and infer beam losses

Objectives:

1. **Learn about AI:** We will use generative AI to produce operationally useful code on top of the model backend.

2. **Get an idea of Streamlit:** Using the provided notebook – you will be giving your LLMs instructions so you need to know what Streamlit is and how it works.

3. **In groups of 2+** (we will allocate groups based on your preference of **project**) work on your tasks. Don't forget to talk to other groups and communicate how you need to share data between groups.

4. **Final Goal: provide tools for machine physicists to predict beam behaviour in the ISIS synchrotron. Explore and define the foundation for future operational tools.** Don't worry if you don't "finish". **You are doing real work, not an exercise.**

Health & Safety

Reference for your week @ RAL

Risk Assessment

- Electrical Equipment:
  - Please be sensible; don't daisy-chain power extensions etc
  - Please report defective equipment to supervisor(s)
- Lifting heavy objects:
  - Do not lift heavy objects, if we have to rearrange furniture please do so in groups
- Fire hazards:
  - Please keep all doorways etc clear of obstructions. Follow fire exits on continuous fire bell. Pulsed fire bell for < 30 seconds = testing
- Trips and Falls:
  - Please report any spillages that may cause slips. Please don't obstruct walkways etc with bags / cables / furniture.
- Regular Computer Use:
  - See next slide

STFC SHE Information

Are you sitting comfortably?

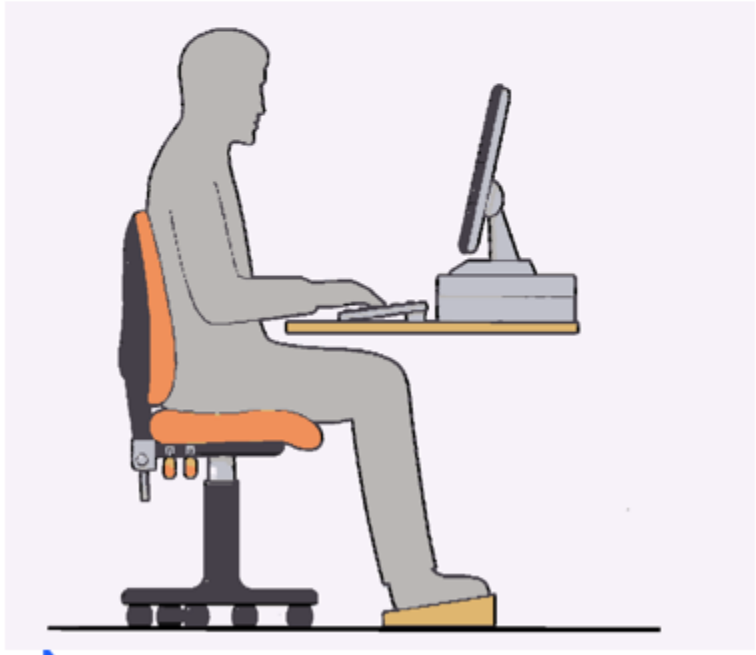
Poor posture can cause backache, upper limb problems & eye strain

Check your posture now!

START

1. Adjust chair height until forearms are at right angles to upper arms.

2. Adjust backrest's height to support lower back



3. Hips should touch the backrest with a gap of 2-4 fingers from edge of seat to back of calf.

4. Adjust monitor height so its top is at eye level, and about an arms length away. Should not have to twist to view screen(s).

5. Sit close to desk (elbows under shoulders) and keep keyboard and mouse close.

6. Feet should be flat on floor or on foot rest to give support and relieve pressure on thighs or backs of the knee.

7. Position monitor at right angles to windows to avoid glare/reflections or close blinds.

FINISH

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a. Sit up straight and deep in the seat. Feet should be flat on the floor or on a footrest;
b. Keep lower arms level with the desk and wrists straight.
c. Sit close enough to your keyboard to eliminate stretching, but far enough to avoid leaning. Your shoulders should be back and your head should be directly over your shoulders;
d. Tap the keyboard lightly; don't strike the keys;
e. Place the mouse within easy reach of the dominant hand. Hold the mouse loosely;
f. Place the monitor at eye level, 16 to 24 inches away;
g. Take short stretch breaks every 20 minutes, as described in the Supervisor Handbook;
h. Exercise eyes frequently. Look away and focus on distant objects; and periodically look up at the ceiling to give your posture muscles a break.

Important Information

What is expected of you

- Responsibility
  - **You are not in school**. This is your chance to **be a researcher**.
  - We are your managers. With respect to work we will give you challenges, you will need to work together, independently, and with us to progress.
  - Please don’t suffer in silence. We are here to do a real project, get help when you need it!
  - You do not magically turn into an adult / grown-up / responsible person etc. at a given age. You are provided opportunities to show yourself that you are capable. This is only such an opportunity if you wish it to be.
  - The more you engage with other humans the more opportunities for growth will be provided.
  - **You are a guest in a scientific laboratory, please do not forget to stay aware of unforeseen hazards. Look after yourself first so that you may look after others.**
  - Worth storing in your phone just in case: RAL Security: +441235 77 8888

What is expected of you

- Working hours
  - **09:00 – 17:00 is our core day**
  - Please **enter via main reception and meet in the Visitors Centre** each morning
  - Don’t worry if you’re late, but please inform us by emailing ralworkexp@stfc.ac.uk in the case of significant delay (> 20 minutes)
  - If you have to stay late on any day (>10 minutes) please notify staff as soon as you are aware
  - 1 hour lunch break; usually 12:00 – 13:00
  - If you cannot attend on any day, please notify staff (and email ralworkexp@stfc.ac.uk) as soon as you are aware (unless you have already done so)
- General notice
  - Although we endeavour to provide and test everything that you need for this week, it is likely we will encounter issues. **This is real work, there will be real problems**.
  - **Don’t suffer in silence**, the sooner we know the sooner we can fix things. **We’re here to help**. Your patience is appreciated.
  - You always have permission to try something interesting, try to keep it within the context of your projects. Use your time wisely.

What is expected of you

- Teamwork:
  - ISIS requires ~150 staff to operate its accelerators, ~650 including everyone. CERN staff ~ 3,000. Humans have enormous potential, but no one is master of all. We must be willing to help one another and work with each other if we are to push the boundaries of knowledge.
- Respect:
  - Working with others often means working with those of different creeds, ideologies, beliefs etc. **If we cannot respect one another's rights to have individual opinions and beliefs, we have little hope of solving issues on the scale of humanity.** Solving such problems starts with respect between individuals.
- Time:
  - We have a limited time to share this experience with you. Please use everyone's time wisely.
  - Some of your time should be spent:
    - Talking to each other and getting to know your teammates / short-term colleagues
    - On the things you are interested in. Learning/trying new things. Investigating interesting options.
    - On project work. Developing software. Obtaining feedback on your work.
    - On final presentations and CV skills.
    - Taking screen breaks. Taking short walks (please don't go too far). Resting body and mind.
    - Thinking. Debating. Sharing and gaining new perspectives.
    - Asking for help / feedback / guidance.
    - Talking to staff about their journeys; take advantage of this opportunity if you have questions about careers, studies, your future…

Housekeeping

- Laptops
  - **Please don't leave site with laptops**. Return them to the charging station when not in use (lunchtime / end of day)
- Location
  - Visitors Centre : Main Room / Hamilton Room is our base for the week
  - During group work sessions you are permitted to work in the outdoor meeting rooms and benches / grass immediately surrounding the Visitors Centre for a maximum of 30 minutes
  - During group work sessions you are permitted to work in the main visitors centre area on the condition that it is not in use
  - If you get lost please locate a staff member and ask them for assistance
- Toilets:
  - Feel free to use at any time when working in the Visitors Centre. **You do not have to ask**.
  - For the ISIS Tour on Monday please ask your guide, preferably use the Visitors Centre facilities before the tour
- Fire Exits:
  - Visitors Centre: main exit / side exit / main room 2 fire escapes
  - Meet on grass behind Visitors Centre (where 2 the outdoor meeting rooms are)
- Emails to Staff:
  - Use the teams channel to contact us
  - In general, please ask in person to save time
  - If not possible, **email** ralworkexp@stfc.ac.uk

Timetable

Date	20.07.26	21.07.26	22.07.26	23.07.26	24.07.26	
Day	**M**	**T**	**W**	**T**	**F**	**Legend**
09:00	**PE Intro**	Project Work	Project Work	Project Work	Finish & Practice Presentations	Staff
09:30	Ice breaker					
10:00	**Handbook pg 9**					
10:30	Laptops / Logins / Prep					
11:00			**Staff Career Presentations**	**Dr Sarah Connors** Climate Scientist @ ESA		WX Students
11:30						
12:00	Lunch	Lunch	Lunch	Lunch	Lunch	Group Work
13:00	Safe AI Usage (Mat et. al.)	**ISIS Tour**	Project Work	Presentation Skills	Prep time	Guest / Tour
13:30					**Final Presentations & Celebration**	
14:00	Project Work					
14:30		Project Work				
15:00				Prepare Presentations		
15:30						
16:00	**Handbook 8/10**	**Handbook 10/15**	**Handbook 11**		**Handbook 12/13/16**	
16:30	Project Work	Project Work	Project Work			

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Let us begin

Getting Started

- Collect a **laptop** from the laptop trolley (using the same one every day is preferred)
- Get your **FedID** and **password** from staff
- ***[old instructions kept just in case]*** Laptop Bitlocker Pin (blue screen on startup) = 89020## where ## are the last 2 numbers of your laptop number, something like 'RLVISCEN020' on a red sticker. Here the last 2 numbers are 20, and the Bitlocker Pin = 809020
- **Before logging in** to your laptop for the first time (and maybe afterwards) **sign in to Eduroam Wifi**
  - User: <your FedID>@stfc.ac.uk
  - Password: your FedID password
- Next **log in to your laptop** with:
  - User: <your FedID>
  - Password: your FedID password
  - This may take aaaaaages to work the first time – please be patient ☺
- **Install** if not already on your laptop (use a browser to download from the web):
  - Visual Studio Code: https://code.visualstudio.com/download
  - JupyterLab via Anaconda: https://www.anaconda.com/download

Setup STFC GitLab account

- Go to https://gitlab.stfc.ac.uk/users/sign_in
- Sign in with your FedID and password
- Follow the instructions to setup two-factor authentication (if prompted)
- Let us know if you have any issues

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  - User: <your FedID>@stfc.ac.uk
  - Password: your FedID password
- Next **log in to your laptop** with:
  - User: <your FedID>
  - Password: your FedID password
  - This may take aaaaaages to work the first time – please be patient ☺
- **Install** if not already on your laptop (use a browser to download from the web):
  - Visual Studio Code: https://code.visualstudio.com/download
  - JupyterLab via Anaconda: https://www.anaconda.com/download

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Optional: Become familiar with Git (spend max 1 hour on this)

- READ the online Git course (up to and including section 9: conflicts) here: https://swcarpentry.github.io/git-novice/
- If you are not familiar with linux / BASH commands please use this reference: https://towardsdatascience.com/basics-of-bash-for-beginners-92e53a4c117a

- Things useful to learn from the Git course:
  - Git clone (copy a repository to a destination such as your PC)
  - Git branch (create your own branch of the repository)
  - Git pull (update your local repository with changes from collaborators)
  - Git status (check which files are tracked and if changes have been made)
  - Git add (add a file/change to be tracked by Git)
  - Git commit (commit added changes to your branch)
  - Git push (send your work to the main repository)
- **Note**:
  - This course will ask you to create a **GitHub** (different from STFC **GitLab**) account – GitHub is free and public. Your STFC GitLab account is not free or public, you will lose access after a few weeks.
  - Please do all work in your STFC GitLab. You are welcome to copy your work to GitHub. We suggest any copying to GitHub is done after your work experience is complete to avoid confusion / issues. Let us know if you'd like help with this.

Project Work

1. Get **corrector dipole** and **trim quad / tune** settings from EPICS (archiver), provide this to your colleagues in a format compatible with the ISIS synchrotron model backend
2. Make a Streamlit GUI that runs the ISIS synchrotron model backend, using EPICS settings, to **calculate, plot, and correct the beam trajectory** (closed orbit)
3. Make a Streamlit GUI that runs the ISIS synchrotron model backend, using EPICS settings, **to plot the beam envelope, with orbit and aperture consideration** (closed orbit plus focussing)
4. Make a Streamlit GUI that runs the ISIS synchrotron model backend, using EPICS settings, **to plot the beam tune (working point)** parameters (related to beam focussing)

ISIS Synchrotron model backend: https://github.com/HaroonRafique/optics_gui_wx26

Everything you need should be in the notebook: https://github.com/HaroonRafique/optics_gui_wx26/blob/main/Dev/12_IO/package_student_guide.ipynb

Tour Tomorrow

13:00 – 14:30: ISIS Target Station 2





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Thanks for your attention, questions?



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Your Presentations (13:30 – 15:30 Friday)

Individual: 3 Slides each

1. Introduce yourself
   - Name, what are you studying?
   - What are your future plans?
   - Pets ☺, Hobbies etc
   - What were your goals for work experience (workbook page 9)
2. Work experience week
   - What did you do (workbook pages 8, 10, 11, 12)?
   - Teamwork?
   - What was fun / difficult / annoying / interesting?
3. Reflections
   - What have you learnt (workbook pages 10 - 13)?
   - How will this experience impact your future (workbook page 16)?

Groups

- Demonstration of your GUIs/work in groups:
  1. Get parameters from EPICS / Archiver
  2. Beam Orbit GUI
  3. Beam Envelope GUI
  4. Beam Working Point GUI
- Demonstrate all of the features and talk through what it does and how it works

Finish presentations with:
- Thank you, any questions?

Handover Instructions

Well done on your excellent contribution! Please make sure that it isn't lost:

- Copy **everything** you've done into a folder of your name / group name / GUI name
  - Include all **python** files
  - Please make sure nothing is lost (input files, include files, etc)
  - If you have work on a **repository** – add a simple text file with the URL of the repository to your files
- Paste onto one of my USB sticks

Develop transverse beam dynamics optimisation tool 1. Build benchmarked model including correctors, trims, and inferred orbits. 2. Develop GUI to visualise beam trajectory and envelope vs aperture. 3. Integrate diagnostics and controls for live feedback and optimisation.	1. Model benchmarked with trims, correctors, and inferred orbits. 2. Python GUI deployed showing trajectory and envelope vs aperture. 3. Integrated tool tested with live diagnostics and controls.	1. Oct 2025 2. Jul 2026 3. Jan 2027

Please save your work for yourself if you want to keep it
Your access to STFC GitLab will be revoked after this week